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Hyperspectral imagery to monitor crop nutrient status within and across growing seasons

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Abstract

Imaging spectroscopy provides the opportunity to monitor nutrient status of vegetation. In crops, prior studies have generally been limited in scope, either to a small wavelength range (e.g., 400-1300 nm), a small number of crop cultivars, a single growth stage or single growing season. Methods that are not time- or site-specific are needed to use imaging spectroscopy for routine monitoring of crop status. Using data from four cultivars of potatoes (*Solanum tuberosum* L.), three growth stages and two growing seasons, we demonstrate the capacity of full-range (400-2350 nm) imaging spectroscopy to quantify nutrient status (petiole nitrate, whole leaf and vine total nitrogen) and predict tuber yield in potatoes across cultivars, growth stages and growing seasons. We specifically tested the capabilities of: (1) ordinary least-squares regression (OLSR) using traditional hyperspectral vegetation indices (VIs); (2) partial least-squares regression (PLSR) using full spectrum (400-2350 nm), VNIR- (visible-to-near infrared: 400-1300 nm) or SWIR-only (shortwave infrared: 1400-2350 nm) wavelengths; (3) predictive models developed for one potato type or planting season on withheld data from a different type or season. Our results show that OLSR models produced poor predictions with data from all dates pooled together (validation R^2 <0.01). Single-date OLSR models performed better (R^2=0.20-0.60,

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relative RMSE=15-30%). PLSR models performed well and were comparable using different spectral regions (full-spectrum, VNIR-only and SWIR-only), with validation R^2=0.68-0.82 and RRMSE=12-25%. Testing across potato types, models produced reliable predictions (R^2=0.45-0.75, RRMSE=13-30%), but with some bias. Cross-season models had validation R^2=0.46-0.75 and RRMSE=17-100%, with a more significant bias than the cross-potato type models. To achieve models that are generalizable and robust, we recommend: (1) obtaining ground measurements that capture the full range of plant growth conditions and developmental stages, and (2) ensuring that image processing approaches minimize spectral discrepancies among dates.

Keywords

Imaging spectroscopy; Petiole nitrate; Foliar nitrogen; Tuber yield; Potato

1. Introduction

Hyperspectral remote sensing has emerged as an important tool to quantify the biochemistry of vegetation (Asner et al., 2015; Singh et al., 2015; Wang et al., 2020) and thus has great potential to monitor nutrient status of crops (He et al., 2016; Mahajan et al., 2017, 2014; Moharana and Dutta, 2016; Pandey et al., 2017; Tremblay et al., 2011), especially given the number of spaceborne imaging spectroscopy missions planned for the 2020's, such as EnMap (Guanter et al., 2015), PRISMA (Galeazzi et al., 2008), HISUI (Matsunaga et al., 2013), SBG (Lee et al., 2015) and CHIME (Nieke and Rast, 2018). Previous research using imaging spectroscopy of plants has primarily focused on natural vegetation, and findings may not be directly applicable to crops that are intensively managed, highly dynamic during a growing season, and especially sensitive to year-to-year environmental variation. As well, agricultural plant traits and nutrient dynamics are often species- or even cultivar-specific, requiring tuning of remote sensing applications based on context.

At present, most of our assumptions about the capabilities of imaging spectroscopy in croplands have been deduced from a small amount of aircraft/UAV (Unmanned Aerial Vehicle) data (Franceschini et al., 2019; Li et al., 2020; Nigon, 2012), or based on leaf-level or proximal measurements using portable spectrometers (e.g., Bohman et al., 2019; Clevers and Kooistra, 2012; Herrmann et al., 2011; Morier et al., 2015; Ray et al., 2006). These studies have generally been limited to one or a small number of cultivars, a single growth stage or a single growing season. One of the biggest hurdles to the use of both air- and spaceborne imaging spectroscopy to quantify and map crop status is the development of approaches that are robust across a wide range of conditions and are not time- (i.e., growth stage), variety- or site-specific. While it may be impossible to design an experiment to capture all dimensions of variation affecting a crop (environment/soils/weather, variety/cultivar, nutrient/water management, pest management, etc.), here we address the development of generalizable imaging spectroscopy mapping approaches using a two-year airborne study of four cultivars of potatoes (*Solanum tuberosum* L.) representing two potato types and experiencing a wide range of nitrogen managements regimes. This provides a foundation to both develop analysis approaches that are repeatable with new data and identify limitations for nutrient monitoring in crops.

Potatoes are the most consumed vegetable crop in the U.S. and the third most important food crop in the world after rice and wheat (Devaux et al., 2020). Plant nitrogen (N) status is critical to the production and quality of potatoes. Severe N deficiency causes premature leaf senescence and decreases tuber yield (Westermann and Kleinkopf, 1985), making fertilizer application and subsequent crop monitoring an important activity for growers. Over-application of N is also detrimental. Excess N application before tuber formation contributes to excessive vine growth and can delay tuber growth (Alva, 2004). In the maturing stage, overfertilization delays tuber maturity and reduces tuber starch content (Alva, 2004). Over-supply of N also has adverse environmental consequences. With heavy rainfall, unincorporated N in the soil can leach into the groundwater or be transported via shallow throughflow to receiving streams, and this causes

groundwater and downstream water quality issues (Errebhi et al., 1998b; Kraft and Stites, 2003; Waddell et al., 2000).

Quantifying crop nutrients – both uptake and availability – is a central component of yield predictions. In recent decades, over 30 models have been developed for predicting potato tuber yield (Raymundo et al., 2014). Using weather, soil characteristics and plant trait data as inputs, these models simulate potato yield under different management practices (e.g., N fertilization, irrigation management) and environmental conditions (e.g., frost, elevated CO_2). To function well, these models require measurements of potato plant properties and responses to environmental variation as inputs (Raymundo et al., 2017). As such, season-long monitoring of potato nutrient status is necessary for both crop management and yield prediction.

Informed N fertilization decisions require timely monitoring of plant N status. Plant tissue analysis has been widely used for this purpose across most crop types. In potatoes, many studies have used petiole nitrate (NO_3-N) on a dry matter basis to guide N applications (Goffart et al., 2008; Haase et al., 2000; Haverkort and MacKerron, 1999; Muñoz-Huerta et al., 2013). Petioles are one tissue where nitrogen is transported and stored in the form of nitrate (NO_3-N), and therefore petiole chemistry indicates the immediate nutritional status of plants and whether they are taking up sufficient nutrients to meet production goals (Millard and Marshall, 1986; Rosen and Bierman, 2008). When petiole nitrate concentration falls below an optimum range, supplemental application of nitrogen fertilizer is recommended (Colquhoun et al., 2019; Rosen and Bierman, 2008). Drawbacks of petiole nitrate for monitoring crop N status include: it is time consuming and labor intensive to measure; it can be highly variable in time and among plants due to different tuber bulking rates and N uptake of the crops associated with different growing conditions when petiole samples are collected, or due to variability of soil conditions in the field; and sometimes responsive N applications based on petiole nitrate analysis may promote vine growth instead of tuber growth. Some researchers have directly measured the NO_3-N concentration in petiole sap (Errebhi et al., 1998a; MacKerron et al., 1995; Vitosh and Silva,

1996; Westcott et al., 1993; Zhang et al., 1996). Although sap petiole NO_3-N correlated well with dry petiole NO_3-N (r > 0.9), regression models were affected by the growth stage at time of sampling, potato cultivars, soil conditions and even the sensors used to measure petiole NO_3-N. In addition, this method is most effective in irrigated areas where the readings are relatively stable due to the homogeneous water status of the plants (Rosen and Bierman, 2008). A comprehensive summary of the protocol of petiole tissue analysis can be found in Haase et al. (2000).

Destructive analysis of other plant organs, such as whole leaves (including petioles and leaflets) and vines (including petioles, leaflets, stems, and flowers), has also been used to diagnose plant N status (Cohen et al., 2010; Herrmann et al., 2010; Reis and Monnerat, 2000; Vos and van der Putten, 1998). In contrast to petioles, whole leaves and vines comprise the tissues where a greater percentage of N is accumulated as amino acids. Therefore, they better indicate the cumulative nutritional status of plants up to the present point in time (Millard, 1988; Rosen and Bierman, 2008).

Non-destructive techniques have been used to measure potato N status indirectly using variables such as leaf/canopy reflectance (Herrmann et al., 2010; Jain et al., 2007; Jongschaap and Booij, 2004; van Evert et al., 2012; Vos and Bom, 1993; Z. Zhou et al., 2018), leaf area index (LAI) (Houlès et al., 2007; Jongschaap and Booij, 2004; van Delden, 2001; Westermann and Kleinkopf, 1985) and leaf chlorophyll content (Gianquinto et al., 2004; Minotti et al., 1994; Olivier et al., 2006; Vos and Bom, 1993; Wu et al., 2007). These variables can be easily measured by ground-based equipment such as the CropScan multispectral radiometer (CROPSCAN, Inc., Rochester, USA), ASD spectroradiometer (Analytical Spectral Devices, Inc., Boulder, USA), LAI-2000 canopy analyzer (LI-COR Biosciences, Lincoln, USA), and SPAD chlorophyll meter (Spectrum Technologies, Inc., Aurora, USA). In general, these instruments utilize the optical properties of leaves or canopies to estimate relevant plant properties, in particular near infrared vs. red reflectance (an indicator of overall biomass greenness, e.g.

128 SPAD and CropScan), transmittance (an indicator of total leaf area, e.g. LAI-2000), or leaf
129 spectra (an indicator of chemical composition, e.g. ASD). For more details about the
130 measurement principles of these sensors, please refer to (Booij et al., 2000; Goffart et al., 2008;
131 Muñoz-Huerta et al., 2013).
132 Non-destructive techniques are being used to infer N status of crops in the field because of their
133 portability, quick response and, in some cases, affordability (Booij et al., 2000). However, these
134 approaches have significant limitations. First, few studies link measured parameters to
135 actionable N management (Booij et al., 2000; Olivier et al., 2006), and destructive sampling is
136 often required to calibrate measured variables against N concentration (Muñoz-Huerta et al.,
137 2013). Second, many non-destructive measurements used to assess foliar N status need to be
138 interpreted with care, because they are not specifically responsive to N supply (Booij et al.,
139 2000). Plant diseases, deficiency of other nutrients, and physiological stresses can also cause a
140 reduction in measured variables even when adequate N is available (Muñoz-Huerta et al., 2013).
141 Finally, with the exception of aerial and satellite-based sensors, non-destructive and destructive
142 measurements conducted at field scale usually occur at discrete, widely separated locations,
143 and thus do not necessarily provide spatially explicit information about where supplements are
144 needed.
145 Numerous studies have used remote sensing techniques to evaluate the nutrient status of
146 potatoes. Most studies regressed vegetation indices (VIs) against variables such as LAI
147 (Clevers et al., 2017; Franceschini et al., 2017; Marshall et al., 2016; Thenkabail et al., 2000),
148 leaf chlorophyll (Clevers et al., 2017; Clevers and Gitelson, 2013; Clevers and Kooistra, 2012;
149 Kooistra and Clevers, 2016), leaf N (Clevers and Gitelson, 2013; Cohen et al., 2010; Herrmann
150 et al., 2010; Hunt et al., 2018; Nigon, 2012; Nigon et al., 2014; Wu et al., 2007), and tuber yield
151 (Al-Gaadi et al., 2016; Bala and Islam, 2009; Gómez et al., 2019). The VI approach seeks to
152 identify spectral bands/wavelengths that highlight specific crop properties and suppress the

effects of other factors. For instance, Clevers et al. (2017) showed that the $\frac{TCRI}{OSAVI}$ ratio was a good linear estimator of leaf chlorophyll at 20-m pixel scales. Here, the transformed chlorophyll absorption reflectance index (TCARI) uses the reflectance at 550, 670 and 700 nm to maximize its sensitivity to leaf chlorophyll (Haboudane et al., 2002) while the optimized soil-adjusted vegetation index (OSAVI) uses the reflectance at 800 and 1510 nm to minimize the effects of soil background (Rondeaux et al., 1996). Herrmann et al. (2010) replaced the 670 nm chlorophyll absorption band with the 1510 nm protein/nitrogen absorption band used in traditional VIs such as TCARI (Haboudane et al., 2002), modified chlorophyll absorption in reflectance index (MCARI) (Daughtry, 2000), and normalized ratio index (NRI) (Ferwerda et al., 2005). Their results showed that the performance of VIs in predicting leaf N was greatly improved after band replacement. Where reflectance measurements across many wavelength regions are available (e.g., with hyperspectral data), some studies explored every possible two-band combination (i.e., $\frac{R_1-R_2}{R_1+R_2}$, where R_i is reflectance at wavelength i) and compared them to crop variables to find combinations that performed best (Heiskanen et al., 2013; Marshall et al., 2016; Thenkabail et al., 2000). Le Maire et al. (2004, 2008) provides an overview of the VIs useful for monitoring potatoes, but there is no consensus on which VI is the most valuable predictor for a given crop variable. This is because of differences among crop types, as well as differences in spectral responses associated with different stresses, environments, phenological growth stages, and varietal characteristics. Since VI-based methods only use a small portion (usually 2-3 spectral bands) of the original spectrum, they do not exploit the additional spectral information contained in other wavelengths that could account for these multiple differences.

Full-spectrum (hyperspectral) reflectance from field spectrometry has been used to monitor potatoes at the leaf level in several studies (Cohen et al., 2010; Couture et al., 2018; Gold et al., 2019; Haboudane et al., 2002; Herrmann et al., 2011; Nigon et al., 2015). In contrast to VIs which use measurements at specific, targeted wavelengths, analysis of full-spectrum data

178 entails the use of continuous measurements across a range of wavelengths (e.g., every 10 nm
179 between 400-2500 nm). Methods for using full-spectrum reflectance must address multi-
180 collinearity among spectral bands because standard statistical approaches are prone to over-
181 fitting of predictive models (Grossman et al., 1996; Verrelst et al., 2016a). A number of
182 techniques have been advanced to address collinearity in hyperspectral data (Feilhauer et al.,
183 2015; Verrelst et al., 2016a, 2015). Among these, the longest- and most widely-used approach
184 is partial least-squares regression (PLSR, Wold et al., 1984). PLSR is a computationally efficient
185 approach that fits a linear regression model by projecting predictor variables and response
186 variables to a small set of uncorrelated latent vectors (Haenlein and Kaplan, 2004). In
187 comparative studies, PLSR generally performs better than VI-based methods (Cohen et al.,
188 2010; Gold et al., 2019; Nigon, 2012; Peng et al., 2019) and equivalently to other methods
189 designed to address collinearity (Wang et al., 2019).
190 To our knowledge, hyperspectral imaging using the full visible-to-shortwave infrared spectrum
191 (VSWIR, 400-2500 nm) has not been widely tested for utility in predicting potato N status and
192 yield potential. First, most prior studies have used signals from the visible–near infrared (VNIR:
193 400–1300 nm, Botha et al., 2007; Cohen et al., 2010, 2007; Jain et al., 2007; Morier et al.,
194 2015), and thus have not taken advantage of numerous spectral features in the shortwave
195 infrared (SWIR, 1400–2500 nm) related to leaf biochemistry (especially proteins, ligno-cellulose,
196 phenolics, sugars and starches; Curran, 1989). Second, most studies have focused on
197 chlorophyll content, which is less instructive to crop monitoring than metrics like petiole NO_3-N
198 or leaf/plant N concentration that are widely used for N management in the potato industry (see
199 Supplemental Table S2). Finally, no studies have explored the model transferability across
200 different growth stages/planting seasons or across different potato cultivars.
201 The overall purpose of this research is to evaluate the performance of aerial hyperspectral
202 remote sensing for quantifying and mapping in-season potato N status and predicting final tuber
203 yield. Specifically, we aim to exploit the rich information content in imaging spectroscopy data of

potato canopies that is related to the chemical composition of plants and thus related to plant N status (indicated by petiole NO_3-N, whole leaf N and whole vine N). We also test the model transferability across different growing seasons and potato types. This will provide a basis for techniques applicable to forthcoming spaceborne sensors.

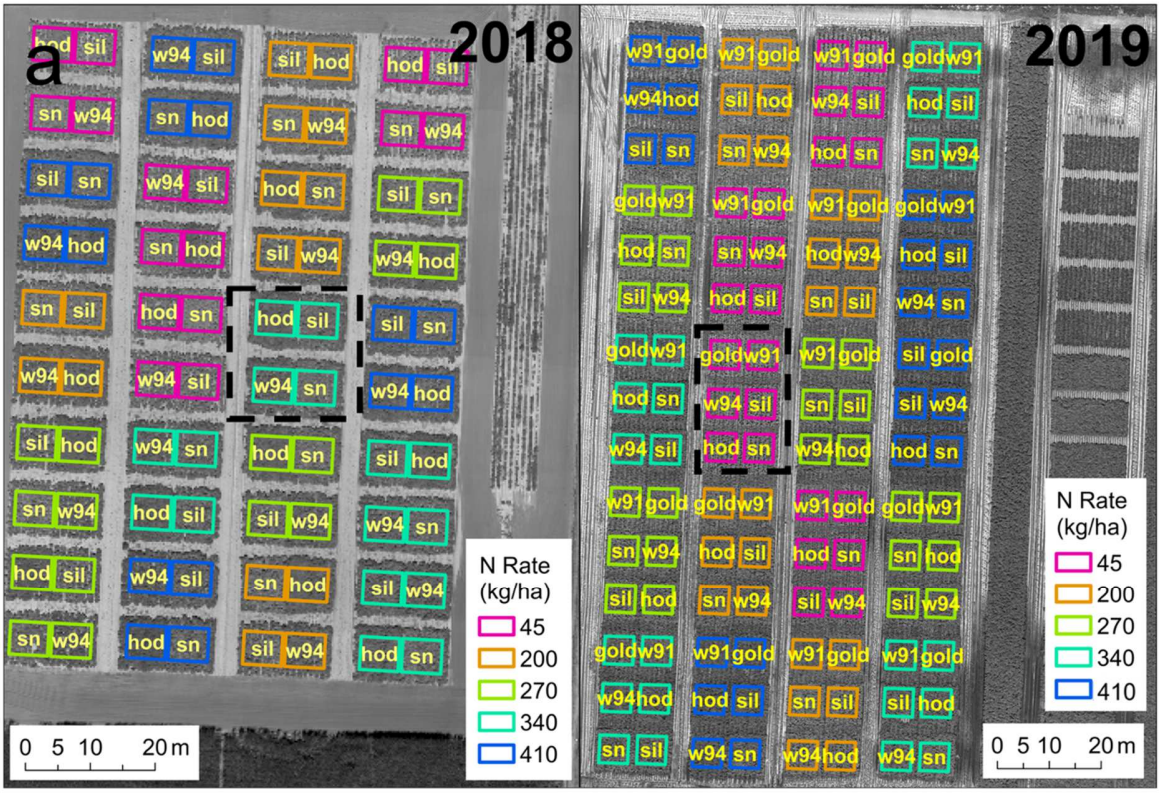
2. Data and Methods

2.1. Field data

Field experiments were conducted in 2018 and 2019 at the University of Wisconsin Hancock Agricultural Research Station (HARS, -89°32'20''E, 44°07'37''N), Hancock, Wisconsin, USA (Figure 1, b). The soils at HARS are a sandy outwash plain characterized by extremely coarse texture (90% sand) with a low soil organic matter content (<10 g kg^{-1}) and a relatively shallow water table (0.9-1.1 m) (Kniffin et al., 2014). The coarse soil texture at Hancock is typical for potato production, together with regular rain events greater than 25 mm in Central Wisconsin and high N needs of potatoes, is a significant contributor to NO_3-N leaching and groundwater contamination (Kniffin et al., 2014). The average temperature during the growing season (May-September) was 19.9 ºC in 2018 and 18.0 ºC in 2019 (data from https://enviroweather.msu.edu/weather.php?stn=hck). Total precipitation during the growing season was 700 mm in 2018 and 617 mm in 2019.

The work was conducted on experimental fields utilizing a split-plot design with four replications (Figure 1, a). Whole plots, such as that enclosed in the black, dashed-line boxes in Figure 1, a, received one of five N application rates including 45, 200, 270, 340, and 410 kg N ha^{-1} total over the growing season. Sub-plots (shown by colored boxes in Figure 1, a) had multiple (four in 2018 and six in 2019) potato cultivars representing two types (or market segments), chipping (Hodag and Snowden in both years) and russet (W9433-1rus and Silverton Russet in both years, and W9133-1rus and Goldrush in 2019 only) (Figure 1, d). The size of each sub-plot was 7.6 m x 7.3 m. Seed tubers were hand planted between 30 April and 4 May of each year. Vine

229 desiccant (Reglone active ingredient diquat dibromide, Syngenta Crop Protection, Greensboro,
230 NC, USA) was applied at a rate of 1.43 L ha^{-1} after diluting 1:2000 with water two weeks before
231 final harvest to promote tuber skin set. Final harvest was conducted using a two-row potato
232 digger (Gallenberg Technologies, Antigo, WI, USA) during the last week of September of each
233 growing season.
234 The timing of fertilizer application was coordinated to five crop growth stages, with different
235 fertilizer rates applied at different stages (Figure 1, c): (1) planting; (2) emergence/hilling; (3)
236 tuber initiation; (4) early tuber bulking; and (5) mid tuber bulking. Other agronomic practices
237 such as planting population, irrigation and pest management were consistent across all rates
238 following local recommendations for potato production (Colquhoun et al., 2018).
239 In-season plant tissues were sampled between 8:00 am and noon (2018: 9 and 16 July; 2019:
240 26 June and 9 July). Here, we only sampled four cultivars: Hodag, Snowden, Silverton Russet
241 and W9433-1rus. Within each sub-plot, petiole and whole leaf samples were collected from the
242 4th leaf from the apex of the shoot on 20 healthy plants. For vine samples, one healthy shoot
243 (including flowers, whole leaves, and stems) was randomly selected from each of six plants, cut
244 into pieces of 5 cm long, and randomly subsampled for 500 g of fresh tissues. All plant tissues
245 were dried in paper bags at 67°C until weight loss ceased, ground to pass through a 2.0 mm
246 screen, and stored in glass vials. Petiole samples were measured for NO_3-N concentration (mg
247 kg^{-1}) using the conductimetric procedure (Carlson et al., 1990). Whole leaf and whole vine
248 samples were measured for total N concentration (mg g^{-1}) using an Elemental Combustion
249 System (Model ECS 4010, Costech, Valencia, CA, USA). Sub-plots of all cultivars were
250 harvested to evaluate tuber yield expressed as Mg ha^{-1}.



b



c

N application (kg/ha)

N Rate	Year	Planting	Hilling	Tuber Initialization	Early Bulking	Middle Bulking
	2018	07-May	24-May	07-June	02-July	16-July
	2019	07-May	29-May	14-June	03-July	18-July
45		45	-	-	-	-
200		45	100	60	-	-
270		45	100	120	-	-
340		45	100	120	70	-
410		45	100	120	70	70

d

Potato cultivars

Marketing Segment	Cultivar (abbreviation)
Chipping	Hodag (hod); Snowden (sn)
Russet	Goldrush (gold); Silverton (sil); W9133-1rus (w91); W9433-1rus (w94)

Figure 1. A) field experimental design (2018: 80 plots = 5 N rates × 4 cultivars × 4 replicates; 2019: 120 = 5 × 6 × 4; black dashed-line boxes show a whole plot containing four or six cultivar subplots). B) the Hancock agricultural research station (HARS) in Wisconsin, USA. C) N application and timing for each of five N rates. D) list of potato cultivars grouped into two potato types (or market segments).

2.2. Remote sensing data

Hyperspectral imagery was collected using co-aligned HySpex (Norsk Elektro Optikk, Norway) VNIR-1800 and SWIR-384 push-broom imagers (Table 1). The VNIR-1800 collects images in 186 spectral bands between 400-1000 nm with a spectral resolution of 3.26 nm. The SWIR-384 provides 288 spectral bands between 953-2518 nm with a resolution of 5.45 nm. An iTraceRT-F400 INS/GNSS sensor (iMAR, Ingbert, Germany) was used to collect IMU (Inertial measurement Unit, e.g., roll, pitch and heading) and GPS (longitude, latitude and altitude) data. The HySpex cameras were flown on a Cessna-180 airplane under clear sky conditions between 11:00 a.m. and 12:30 p.m. local time (acquisition dates: 6 and 18 July 2018; 25 June and 8 July 2019) at an altitude of 220-360 m above the ground, generating 0.15-0.25 m pixels.

The processing of HySpex images included:

(1) sensor boresighting

Forty ground control points and two Hyspex images, were collected at a tennis court in Madison, Wisconsin to estimate the boresighting offsets (i.e., roll, pitch and heading offsets) of the Hyspex cameras. The estimated boresighting offsets were then used to remove the spatial mismatch between the images from the two cameras in step (4). The tennis courts are periodically re-flown and checked against to GCPs during the flight season to ensure quality of the geo-correction.

(2) radiometric calibration

Radiometric calibration was performed to convert image digital numbers (DN) to at-sensor radiance (in mW nm^{-1} sr^{-1} cm^{-2}): $\mathrm{Radiance} = a \times \mathrm{DN} + b$. Radiometric calibration coefficients (a and b) were calculated using the header information (e.g., integration time, quantum efficiency, sensor FOV, spectral wavelength and interval) stored in HySpex DN images.

(3) smile-effect correction

281 “Smile” is the shift in wavelength position from pixel to pixel across the focal array.
282 Correction of smile effects was accomplished by matching the absorption features of
283 radiance spectra with atmospheric O_2, CO_2 and H_2O absorption features at 686, 762,
284 820, 940, 1130, 1268, 1470 1572, 2004 and 2055 nm (Gao et al., 2004; Richter et al.,
285 2011). The estimated shifts were then used to normalize radiance spectra to consistent
286 wavelengths using linear interpolation.
287 (4) geometric correction
288 A ray-tracing algorithm was performed to ortho-rectify radiance images (Meyer, 1994;
289 Schläpfer and Richter, 2002). The boresighting offsets from step (1), on-board IMU and
290 GPS data from the iTraceRT-F400 INS/GNSS sensor, and a 1m-resolution digital
291 elevation model image were used as inputs into the ray-tracing algorithm.
292 (5) atmospheric correction
293 Atmospheric correction was performed to convert at-sensor radiance to surface
294 reflectance (range: 0-100%). Water vapor and visibility were estimated from image
295 spectra (spectra features used for estimating water vapor: 940 and 1300 nm; for
296 estimating visibility: 450 nm) and were used to sequentially remove the atmospheric
297 effects (e.g., scattering and absorption) on surface reflectance following (Kaufman et al.,
298 1997; Kaufman and Gao, 1992; Richter and Schläpfer, 2002).
299 (6) bidirectional reflectance distribution function (BRDF) correction
300 BRDF correction was performed to reduce across-track canopy anisotropic effects
301 caused by the sun-sensor viewing angle geometry (Collings et al., 2010; Schlapfer et al.,
302 2015; Wanner et al., 1995; Weyermann et al., 2015). The Ross-volumetric and Li-thick
303 kernels were used to simulate the anisotropic scattering effects of canopies (Wanner et
304 al., 1995).
305 (7) spectral smoothing and data extraction for statistical analyses

Savitzky–Golay filter (window size: 7) was used to reduce noise in surface reflectance (Schafer, 2011). Then, the average reflectance of all vegetation pixels within each subplot was extracted (Figure 1, a). Non-vegetation pixels (e.g., shadows and soil) were excluded using NDVI<0.5.

(8) vector normalization

Vector normalization was performed to reduce the effects of illumination conditions on reflectance (Asner et al., 2015; Feilhauer et al., 2010). Noisy bands (2350-2500 nm) and bands at wavelengths within the absorption spectrum of atmospheric water vapor (1290-1500 nm, 1790-2050 nm) were excluded from analysis.

All steps were performed in open-source Python (version 3.7.4) packages. The HyPro package (available from https://github.com/EnSpec/HyPro) was used to do the image pre-processing described in steps 1-5 (Liu et al., 2019). Step 6 was performed in the HyTools package (https://github.com/EnSpec/HyTools), which provides a series of processing tools for hyperspectral data analysis such as spectral resampling, topographic correction and BRDF correction (following Wang et al., 2020). Steps 7-8 were performed using the open-source Geospatial Data Abstraction Library (GDAL, version 3.0.4), as well as Python packages Pillow (6.2.0), Numpy (1.17.2) and Scipy (1.3.1).

Table 1. Specifications of HySpex VNIR-1800 and SWIR-384 cameras.

Specifications	**VNIR-1800**	**SWIR-384**
Spectral range (nm)	400-1000	953-2518
FWHM (nm)	3.26	5.45
Number of bands	186	288
Spatial pixels	1800	384
FOV across track (deg)	34	32
IFOV across/along track (mrad)	0.32/0.64	1.46/1.46

FWHM: Full Width at Half Maximum; FOV: Field-Of-View; IFOV: Instantaneous Field-Of-View.

2.3. Statistical models

We built ordinary least-squares regression (OLSR) and PLSR statistical models to predict potato variables as a function of image reflectance. All models, except the cross-season ones (see Section 3.6), were built by pooling data across dates. For tuber yield prediction, we only used the spectral data from the latest image per year (16 July 2018 and 10 July 2019). Although date-specific models would be expected to perform better, they would not be generalizable outside of a given date or the associated developmental stage at that date. The OLSR model linked potato variables (including petiole NO_3-N, whole leaf total N, whole vine total N and tuber yield) with VIs: $y = a \times VI + b$. Here, we calculated the VIs commonly used in previous studies (Table 2, Liu et al., 2017; Thenkabail et al., 2018). These VIs were grouped into three categories based on their sensitivity to foliar biochemistry including: chlorophyll, water and dry matter content.

Compared to OLSR using VIs (Table 2), the PLSR approach offers the potential to make use of the full spectrum for crop assessment. The PLSR model related potato variables with the reflectance at all spectral bands: $y = \sum \alpha_i \times \rho_i + \beta$. PLSR is useful when the number of observations is less than the number of predictors variables (here, the number of spectral bands) (Verrelst et al., 2015). In this study, we established the number of latent factors to use in PLSR as the number that minimized the prediction residual error sum of squares (PRESS) (Chen et al., 2004; Singh et al., 2015; Wang et al., 2019): (1) the original dataset was split into calibration (75%) and validation (25%) datasets; (2) PLSR models with different numbers of latent factors (1-30) were built using the calibration dataset; (3) models were applied to the validation dataset to evaluate PRESS; (4) the number with the lowest PRESS value was chosen as the number of latent factors for PLSR models. The standardized regression coefficients were used to evaluate the contribution of each wavelength to the estimation of potato variables (Singh et al., 2015; Wang et al., 2019; Wold et al., 2001).

We evaluated the performance of PLSR models using three spectral regions: full spectrum (400-2350 nm), VNIR spectrum (400-1300 nm) and SWIR spectrum (1400-2350 nm), excluding water absorption bands. This enables testing whether specific spectral ranges are more robust than others, for example by excluding wavelengths that may confound the analyses. In particular, we wanted to test the use of the SWIR alone, as this is the wavelength range containing the strongest absorption features associated with nitrogen (i.e., proteins) that is not either a) confounded by other leaf constituents with overlapping spectral features, or b) leveraging an imperfect correlation between chlorophyll (which contains nitrogen) and leaf N (Féret et al., 2020).

We also evaluated the performance of PLSR models across different potato types, since potato varieties in each type have been developed to express different traits. The original dataset from all cultivars was first split into two types/groups: chipping (including Hodag and Snowden) and russet (Silverton Russet and W9433-1rus). Then, one potato type was used for calibrating models while the other one was for validating models. Finally, we evaluated the performance of PLSR models across different planting seasons. Similar to the cross-potato type validation, we split the original dataset into two years (2018 and 2019), with one year used for calibration and the other one for validation. For the cross-potato type and cross-season analyses, we only present the results of using full-spectrum data to build models.

Table 2. Hyperspectral vegetation indices used in previous studies. Adapted from Liu et al. (2017) and Thenkabail et al. (2018).

Index	Formula	References
Chlorophyll		
NDVI (Normalized Difference Vegetation Index)	$\frac{\rho_{810} - \rho_{670}}{\rho_{810} + \rho_{670}}$	(Tucker, 1979)
SIPI (Structure Insensitive Pigment Index)	$\frac{\rho_{800} - \rho_{445}}{\rho_{800} + \rho_{680}}$	(Penuelas et al., 1995)

EVI (Enhanced Vegetation Index)	$2.5\frac{\rho_{800}-\rho_{680}}{\rho_{800}+6\rho_{680}-7.5\rho_{450}+1}$	(Huete, 1997)
ARVI (Atmospherically Resistant Vegetation Index)	$\frac{\rho_{800}-2\rho_{680}+\rho_{450}}{\rho_{800}+2\rho_{680}+\rho_{450}}$	(Kaufman and Tanre, 1992)
SAVI (Soil-Adjusted Vegetation Index)	$\frac{1.5(\rho_{800}-\rho_{680})}{\rho_{800}+\rho_{680}+0.5}$	(Huete, 1988)
CARI (Chlorophyll Absorption in Reflectance Index)	$(\rho_{700}-\rho_{670})-0.2(\rho_{700}-\rho_{550})$	(Daughtry, 2000)
$CI_{red\text{-}edge}$ (Chlorophyll index in red edge)	$\frac{\rho_{800}}{\rho_{720}}-1$	(Zarco-Tejada et al., 2002)
MCARI (Modified Chlorophyll Absorption Reflectance Index)	$\frac{[(\rho_{700}-\rho_{670})-0.2(\rho_{700}-\rho_{550})]\rho_{700}}{\rho_{670}}$	(Daughtry, 2000)
TCARI (Transformed Chlorophyll Absorption Reflectance Index)	$3\left[(\rho_{700}-\rho_{670})-\frac{0.2(\rho_{700}-\rho_{550})\rho_{700}}{\rho_{670}}\right]$	(Haboudane et al., 2002)
$PSSR_{a\text{-}c}$ (Pigment-Specific Simple Ratio)	$\frac{\rho_{800}}{\rho_{680}},\frac{\rho_{800}}{\rho_{635}},\frac{\rho_{800}}{\rho_{470}}$	(Blackburn, 1998)
$CRI_{550,700}$ (Carotenoid Concentration Index)	$\frac{1}{\rho_{515}}-\frac{1}{\rho_{550}},\frac{1}{\rho_{515}}-\frac{1}{\rho_{700}}$	(Gitelson et al., 2003)
PRI (Photochemical Reflectance Index)	$\frac{\rho_{570}-\rho_{530}}{\rho_{570}+\rho_{530}}$	(Gamon et al., 1992)
PSRI (Plant Senescent Reflectance Index)	$\frac{\rho_{680}-\rho_{500}}{\rho_{750}}$	(Merzlyak et al., 1999)
CI (Curvature Index)	$\frac{\rho_{675}\rho_{690}}{\rho_{683}^2}$	(Zarco-Tejada et al., 2002)
Water		
NDWI (Normalized Difference Water Index)	$\frac{\rho_{862}-\rho_{1239}}{\rho_{862}+\rho_{1239}}$	(Gao, 1996)

WI (Water Index)	$\frac{\rho_{900}}{\rho_{970}}$	(Penuelas et al., 1997)
MSI (Moisture Stress Index)	$\frac{\rho_{1599}}{\rho_{970}}$	(Hunte and Rock, 1989)
SIWSI (Short Infrared Water Stress Index)	$\frac{\rho_{858} - \rho_{1640}}{\rho_{858} + \rho_{1640}}$	(Fensholt and Sandholt, 2003)
SRWI (Simple Ratio Water Index)	$\frac{\rho_{860}}{\rho_{1240}}$	(Fensholt and Sandholt, 2003)
NDII (Normalized Difference Infrared Index)	$\frac{\rho_{819} - \rho_{1649}}{\rho_{819} + \rho_{1649}}$	(Hardisky et al., 1983)
WBI (Water Band Index)	$\frac{\rho_{900}}{\rho_{970}}$	(Penuelas et al., 1997)
Dry matter		
NDTI (Normalized Difference Tillage Index)	$\frac{\rho_{1650} - \rho_{2215}}{\rho_{1650} + \rho_{2215}}$	(Deventer et al., 1997)
NDMI (Normalized Dry Matter Index)	$\frac{\rho_{1650} - \rho_{1722}}{\rho_{1650} + \rho_{1722}}$	(Wang et al., 2011)
LCA (lignin-Cellulose Absorption Index)	$2\rho_{2205} - (\rho_{2165} + \rho_{2330})$	(Daughtry et al., 2005)

2.4. Model evaluation

A repeated *k*-fold cross-validation (4 folds, 125 repeats) was performed to evaluate models (Figure 2). First, the original dataset was randomly partitioned into calibration (75%) and validation (25%) datasets, with the validation one withheld from all calibration efforts. Then, the calibration dataset was randomly split into four folds/groups, with each fold used as training (3X) and testing (1X) datasets. A model was fit on the training set, and then applied to the testing and withheld validation sets. This step was repeated 125 times, yielding 500 (=4 folds × 125 repeats) model predictions for each data point. The average of 500 model predictions was used

as the model estimation of each data point, and the standard deviation as the estimation uncertainty. The coefficient of determination (R^2), root mean squared error (RMSE) and relative RMSE (RRMSE, RMSE/(Max-Min)) were calculated to evaluate the performance of models on the training, testing and validation datasets (Singh et al., 2015; Wang et al., 2019). The sample size of the training, testing and validation datasets for all analyses can be found in Supplemental Table S 1.

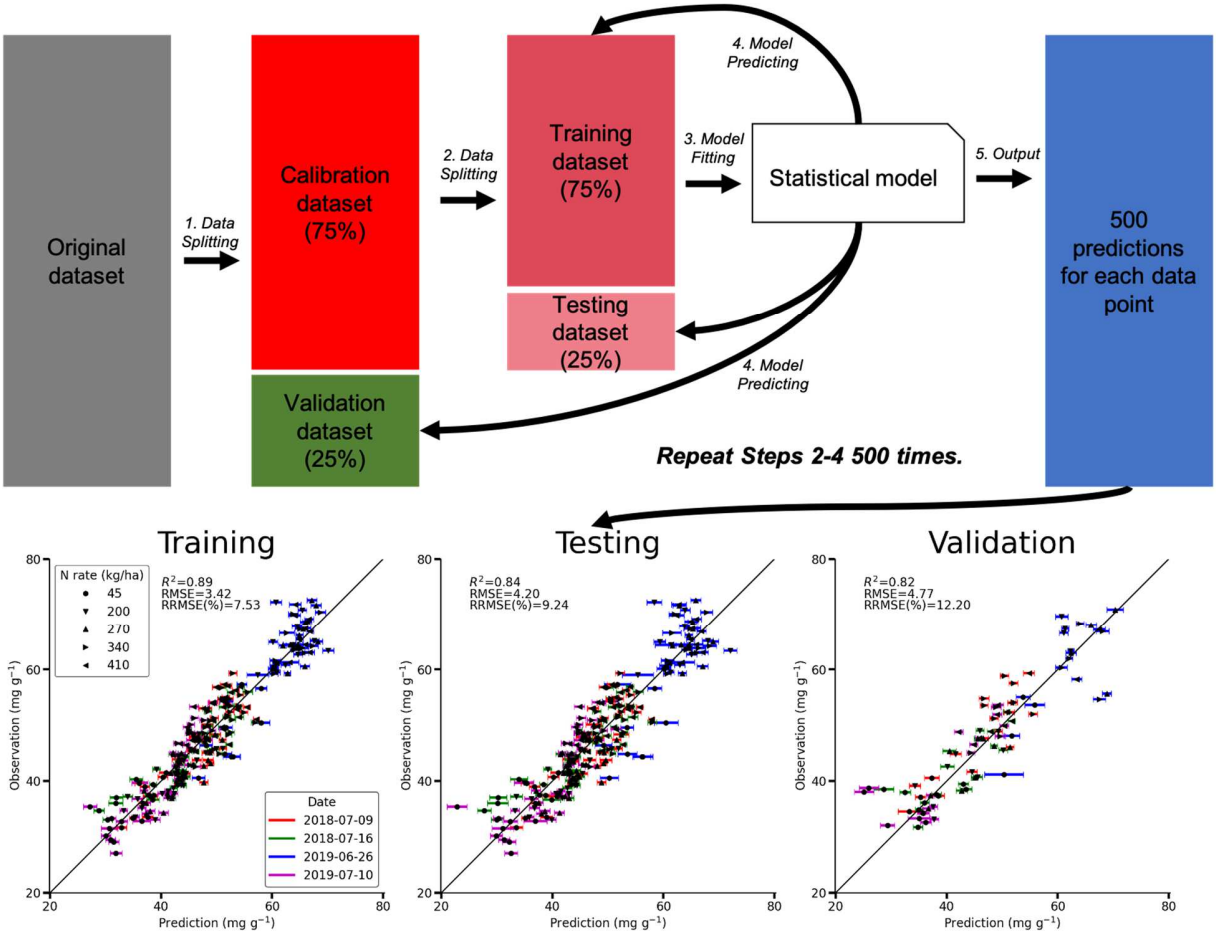


Figure 2. Repeated k-fold cross-validation procedure.

3. Results

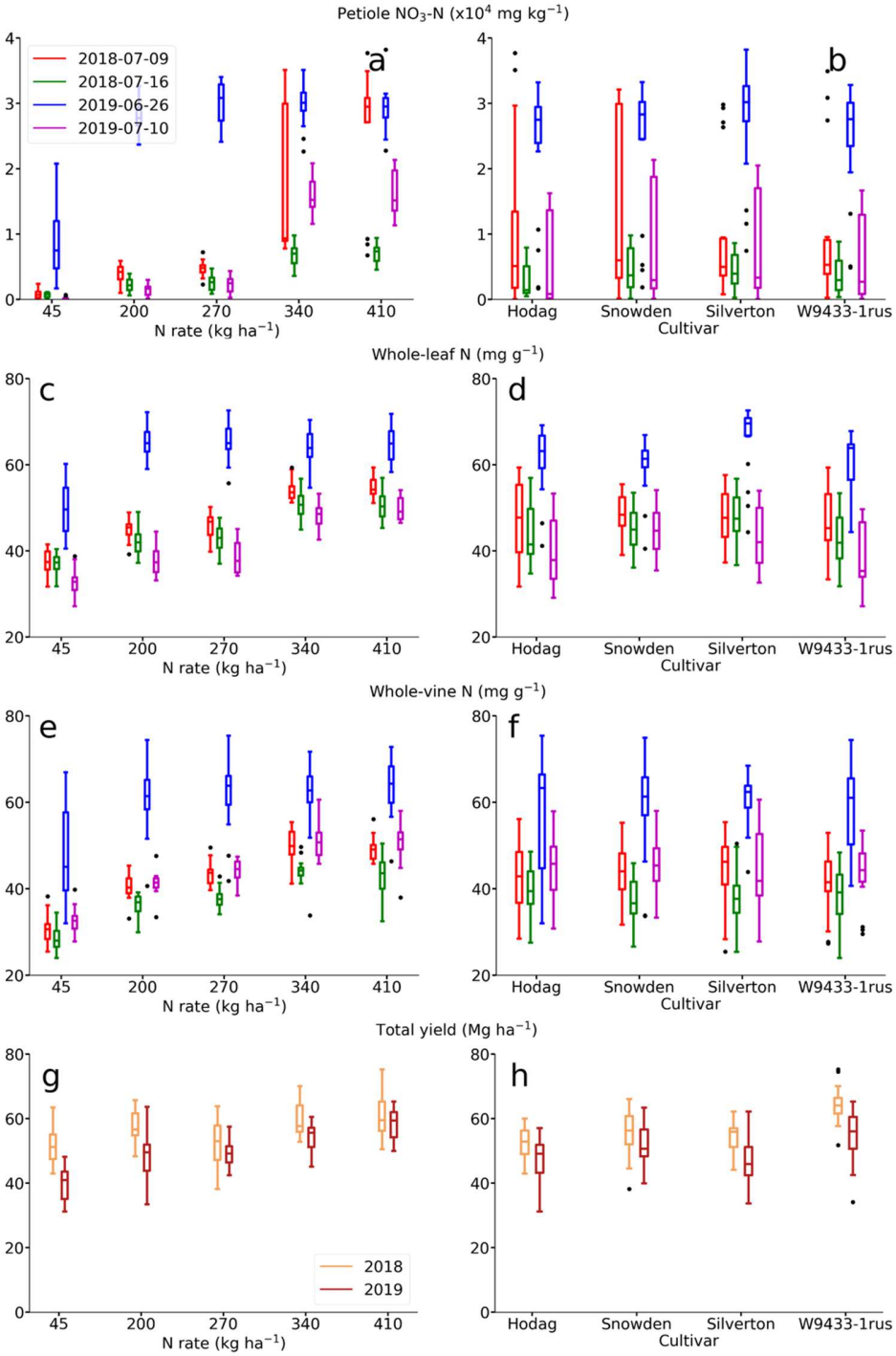
3.1. Plant N concentration and tuber yield

In general, plant N concentration (indicated by petiole NO_3-N, whole leaf N and whole vine N) were closely related to the total N fertilizer applied at the time of sampling (Figure 3, a, c, e). For instance, plots under 200, 270, 340, 410 kg N ha^{-1} received the same N application before tuber initiation (Figure 1, c), and thus exhibited similar N concentration (Figure 3, June 26, 2019). As split applications were applied, plots receiving more N fertilizer showed higher N concentration in plant tissues (Figure 3, July 09, 10 and 16).

Early growth stages (9 July 2018 and 26 June 2019) had higher N concentration than late growth stages (16 July 2018 and 10 July 2019) (two-sample unpaired *t*-test, $p < 0.01$). This reflects the high N demand for tuber bulking and yield gain over the course of the growing season, i.e. the movement of N from foliage to underground tubers.

Tuber yield generally increased with more N applied (Figure 3, g). Notably, yield at 270 kg N ha^{-1} was lower than that at 200 kg N ha^{-1} in 2018. This may have resulted from higher leaching potential at the higher N rate following N application during tuber initiation (Figure 1, c). As for the yield across two seasons, 2018 had a significantly higher yield than 2019 under N rates of 45, 200, 340 kg N ha^{-1} (two-sample unpaired *t*-test, $p < 0.01$), due to lower growing degree days associated with overcast weather during the early stages of tuber growth in 2019 (Section 2.1).

Plant N concentration generally did not show significant differences among the four potato cultivars (Figure 3b, d, f, one-way analysis of variance (ANOVA), $p = 0.50$-0.97). One exception was for the whole leaf total N on 16 July 2018 and 26 June 2019 (Figure 3, d; ANOVA, $p = 0.04$). In contrast, tuber yield significantly differed among cultivars (Figure 3, h; ANOVA, $p < 0.01$), indicating different yield potential of different cultivars.



410

411 Figure 3. Comparison of potato N concentration (including petiole NO_3-N (a, b), whole leaf total N (c, d),
412 whole vine total N (e, f) and tuber yield (g, h) among five N rates (left column) and four potato cultivars
413 (right column).

3.2. Spectral characteristics of potato canopies

In the visible (VIS, 400-690 nm) and shortwave infrared (SWIR, 1400-2350 nm) wavelength regions, reflectance decreased with N rates (Figure 4, a, c, e, g). However, the reflectance in the near infrared (NIR, here 700-1300 nm) region increased with the total N fertilizer applied at time of imaging. This is mainly because high N rates tended to result in higher LAI, thereby enhancing the absorption of chlorophyll, water and dry matter in the VNIR and SWIR, as well as the multiple backscattering of photons in the NIR (Jensen, 2000).

Rates 200 and 270 kg N ha^{-1}, as well as rates 340 and 410 kg N ha^{-1}, exhibited negligible differences (<1%) in reflectance from each other in 2018 (Figure 4, a, c)., despite receiving different amounts of N (Figure 1, c). This spectral reflectance overlap between N rates has also been reported in previous studies (Botha et al., 2007; Jain et al., 2007; Morier et al., 2015). It implies that the change in canopy reflectance may not be detectable when the difference in N application is under a certain threshold.

A temporal change in reflectance was also observed (Figure 4, a, c, e, g). In the VIS and SWIR, the reflectance at late growth stages was about 0.01 lower than that at early stages. In contrast, the temporal change in the NIR reflectance was more apparent: late growth stages (16 July 2018 and 10 July 2019) had reflectance 0.05-0.25 higher than early stages (9 July 2018 and 26 June 2019).

Interestingly, spectral differences were observed among the four cultivars even though they received the same N applications (Figure 4, b, d, f, h). The chipping cultivars (Hodag and Snowden) generally had higher NIR but lower SWIR reflectance than the russet cultivars (Silverton Russet and W9433-1rus). This spectral difference among cultivars did not seem to be related to foliar N concentration (Figure 3, b, d, f), and may have been associated with vegetative cover, chlorophyll content, or genetic factors.

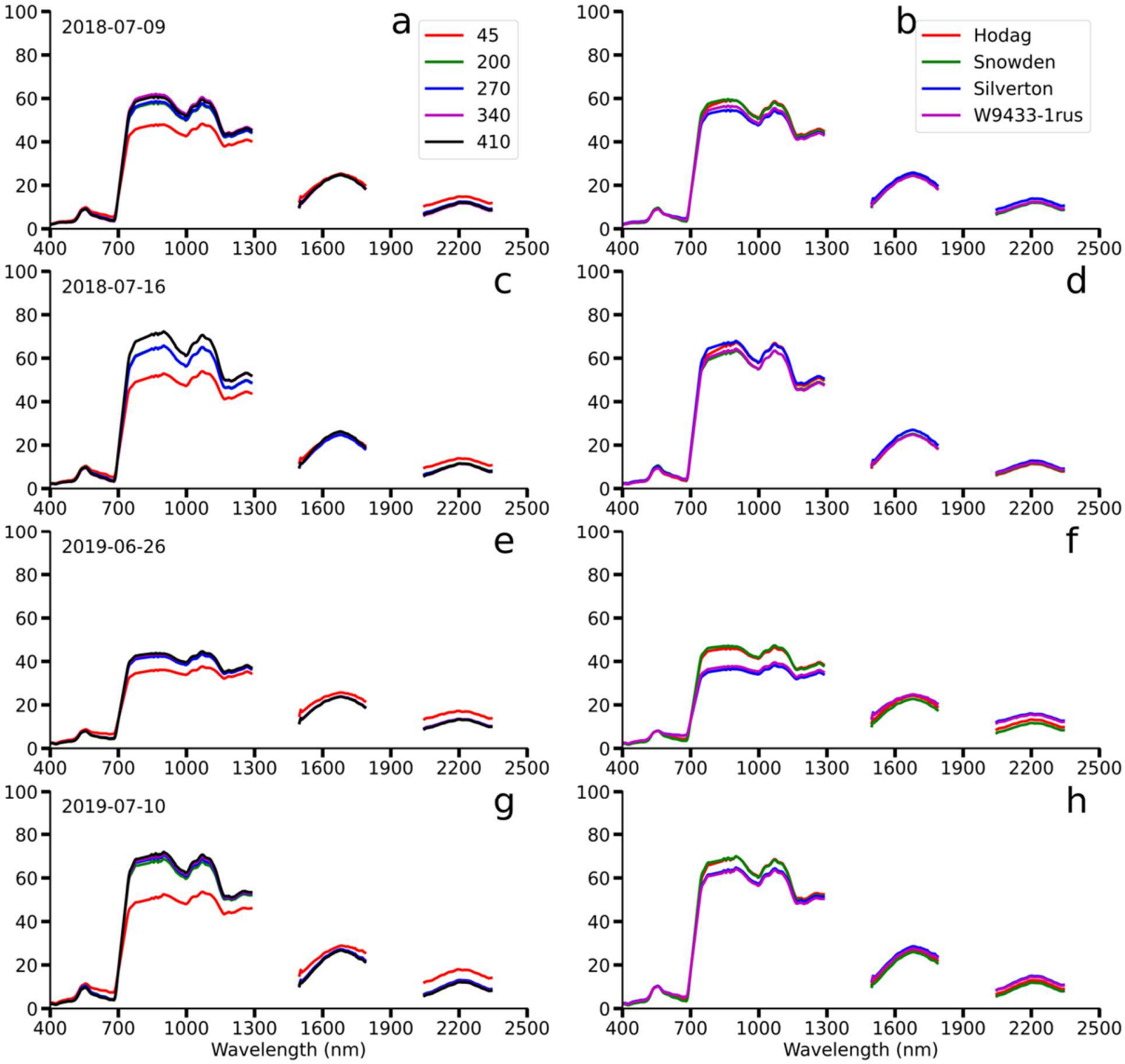
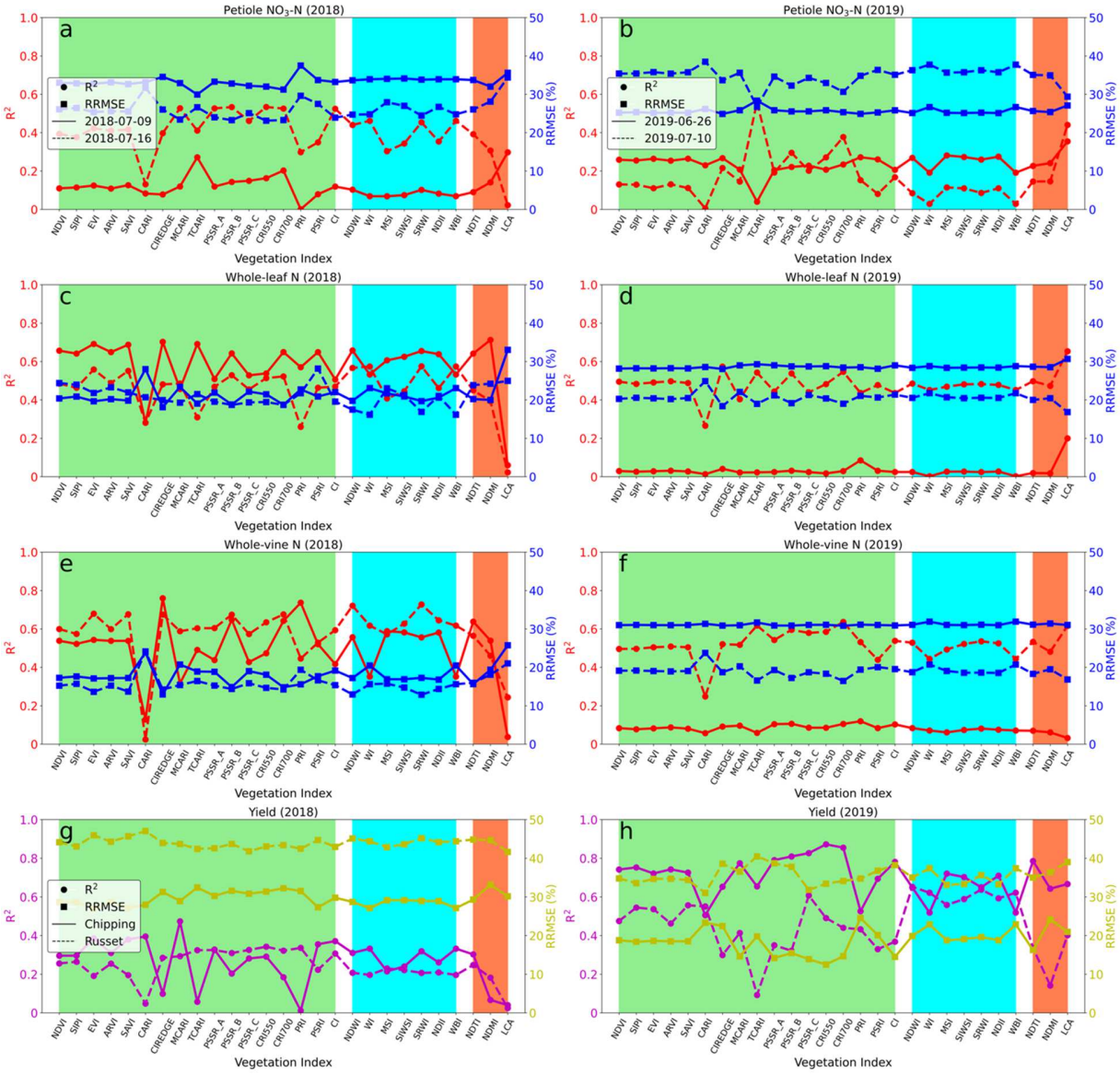


Figure 4. Comparison of mean reflectance (0-100%) among five N rates (left column) and four potato cultivars (right column). Mean reflectance was calculated by averaging the spectra of plots with the same N rate or potato cultivar.

3.3. OLSR modelling

OLSR models generally produced poor predictions for plant tissue N concentration and tuber yield when data from all dates were pooled together (validation $R^2 < 0.01$). Specifically, no single OLSR model produced meaningful predictions on all dates or across years, suggesting that VI-based models could not be generalized (Botha et al., 2007; Cohen et al., 2007; Franceschini et

449 al., 2017; Kooistra and Clevers, 2016). Therefore, we explored the potential of VIs to predict
450 potato variables at each growth stage (Figure 5).
451 Chlorophyll and water VIs use the wavelengths around chlorophyll (550, 680 710, and 800 nm)
452 and water (860, 940 and 1240 nm) absorption features (Table 2). Considering the notable
453 spectral difference in the VIS-NIR among N rates (Figure 4 (a, c, e, g) and Section 3.2), it is
454 suggested that these two groups of VIs produced predictions with reasonable accuracies in
455 most cases (validation R^2 = 0.45-0.60).
456 Dry matter VIs (NDTI, NDMI and LCA) exhibited comparable predictive ability (validation R^2 =
457 0.40-0.60) even though the SWIR reflectance (1650, 1720, 2265 and 2210 nm) used by these
458 VIs had small variations among N rates (Figure 4, a, c, e, g). We also note that the model
459 prediction for 26 June 2019 was relatively poor (validation R^2 < 0.2, Figure 5, right), mainly due
460 to the small variations in plant N concentration (Figure 3, 2019-06-26) and spectral reflectance
461 (Figure 4, e).



462

463 Figure 5. OLSR modelling validation results (left axis, red and purple symbols: R^2; right axis, blue and
464 gold symbols: relative RMSE) of N concentrations including petiole NO_3-N, whole leaf N and whole vine N,
465 and tuber yield for chipping (solid lines) and russet (dash lines) types using different VIs in Table 2.
466 Background colors represent different groups of VIs. Green: chlorophyll-sensitive; Cyan: water; Orange:
467 dry matter.

468

469 **3.4. PLSR modelling with different spectral regions**

470 The model evaluation statistics for all PLSR models (Sections 3.4 and 3.5) developed in this
471 study are reported in Supplemental Table S 1. These results are referenced in each section as
472 we present the results from different subsets of the spectral data.

Full-spectrum models exhibited the best performance, with validation R^2 values ranging from 0.70-0.82 (Figure 6 a, b, c and Figure 7 a, b). The performance of VNIR- and SWIR-spectrum models was comparable to each other, but these models had slightly lower R^2 values (about 0.03-0.10 less) than full-spectrum models (Figure 6 d-i and Figure 7 c-f).

The coefficient curves of full-spectrum models had some broad similarities with VNIR-spectrum models (Figure 8), indicating that both model formulations keyed in on similar features. However, the coefficient curves of full- and VNIR-spectrum models differed from SWIR-spectrum models. When using the full spectrum to build models, the contribution from SWIR wavelengths appeared to be suppressed due to small variations in SWIR reflectance among N rates (Figure 4), and thus the full- and VNIR-spectrum models utilized common spectral features.

Most of the important wavelengths identified from full- (blue bars in Figure 8) and VNIR-spectrum models (green bars) were located around 400-450, 850-900 and 1000-1150 nm. Although some full-spectrum models exhibited important wavelengths in the SWIR known to be related to proteins (Curran, 1989), such as 1500-1575 nm and 2030-2055 nm, they generally did not overlap with the important wavelengths (red bars in Figure 8) found from SWIR-spectrum models (e.g., 1580-1590, 1750-1800 and 2150-2220 nm).

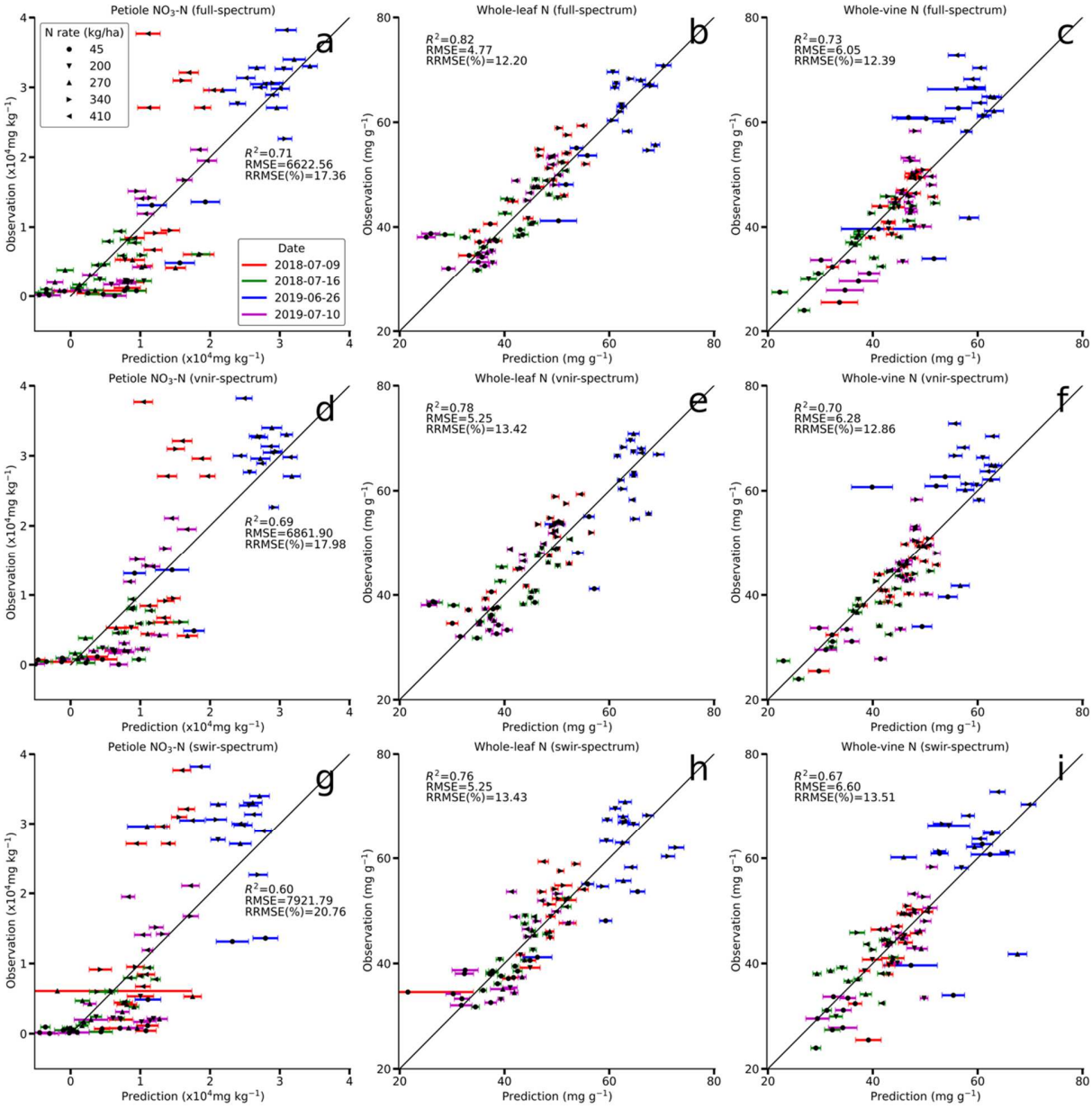
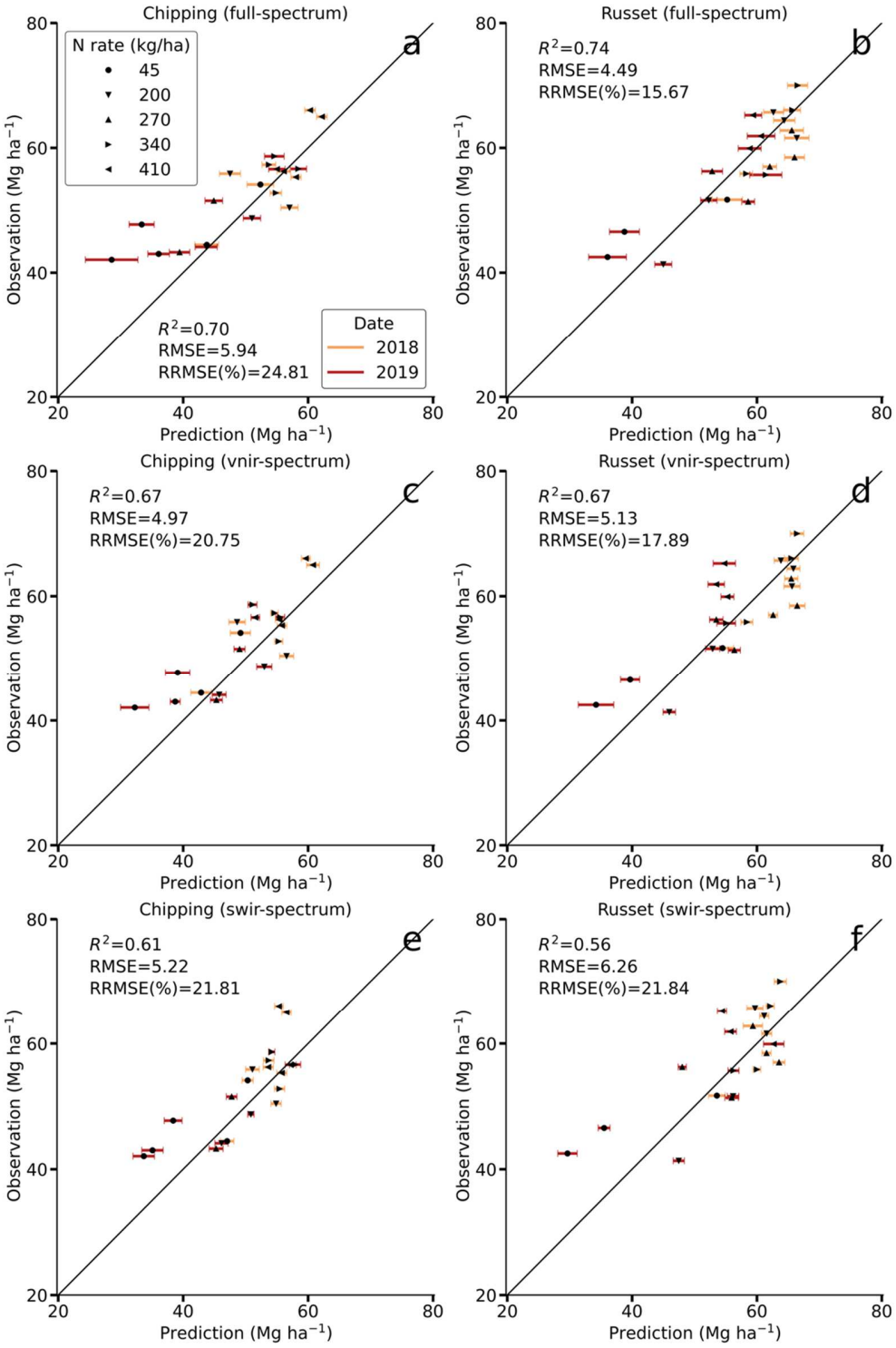


Figure 6. HySpex PLSR modelling validation results of petiole NO_3-N (left column), whole leaf N (middle column), and whole vine N (right column). Notes: (1) horizontal bars indicate the uncertainty (±1 standard deviation) in predictions; (2) full-spectrum: 400-2500 nm (top row); VNIR-spectrum: 400-1300 nm (middle row); SWIR-spectrum: 1400-2500 nm (bottom row).



494

495 Figure 7. HySpex PLSR modelling validation results of tuber yield over two years (left column: chipping
496 type; right: russet type). Notes: (1) horizontal bars indicate the uncertainty (±1 standard deviation) in
497 predictions; (2) full-spectrum: 400-2500 nm; VNIR-spectrum: 400-1300 nm; SWIR-spectrum: 1400-2500
498 nm; (3) spectra from 16 July 2018 and 10 July 2019 are used for yield modelling.

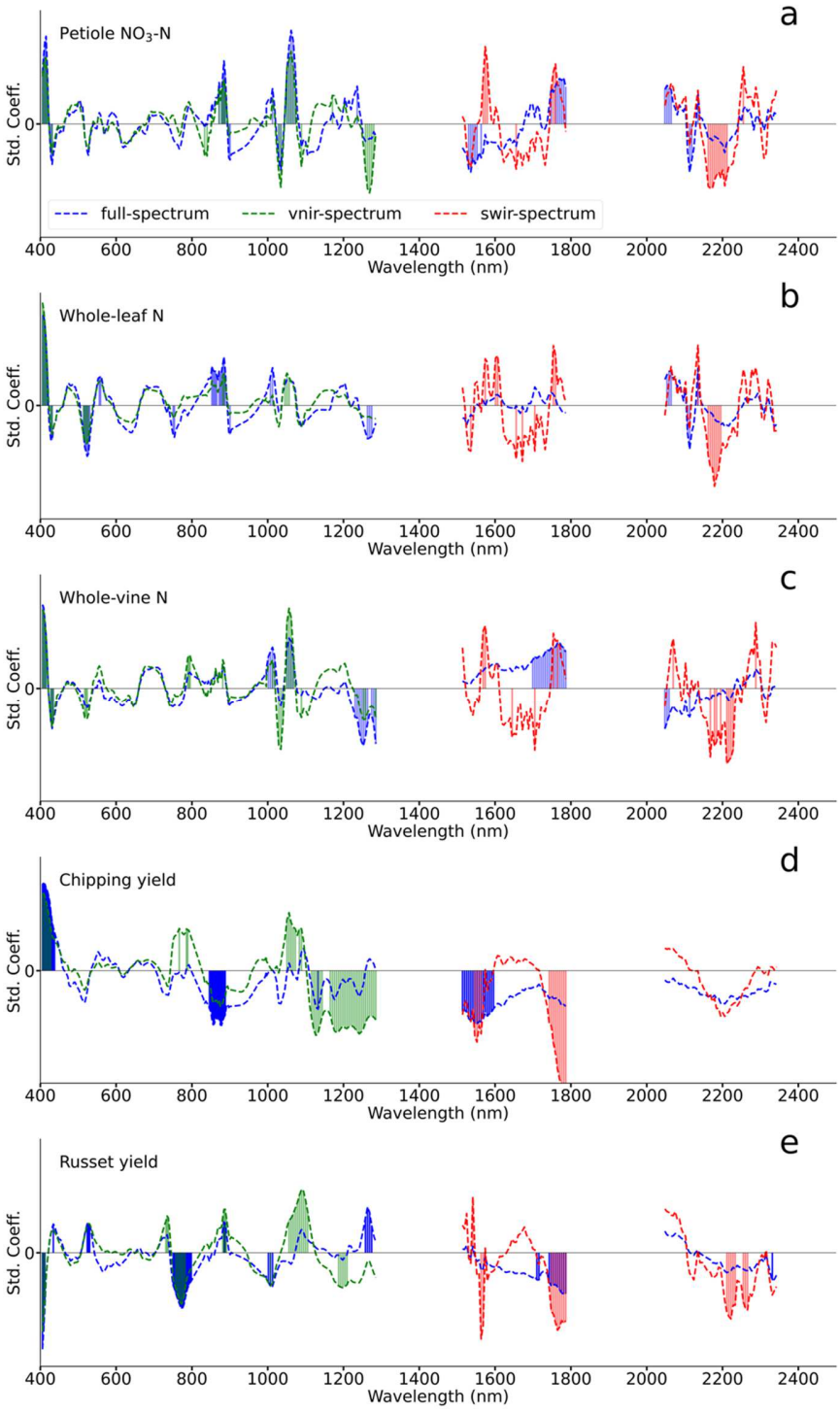


Figure 8. Standardized PLSR coefficients using different spectral wavelength regions (full-spectrum in blue: 400-2500nm; VNIR-spectrum in green: 400-1300nm; SWIR-spectrum in red: 1400-2500nm). Notes: (1) each coefficient curve is scaled by $\frac{1}{\alpha_{max}-\alpha_{min}}$, where α_{min} and α_{max} are the minimal and maximal coefficients; (2) vertical bars represent the locations of wavelengths important for predicting potato variables.

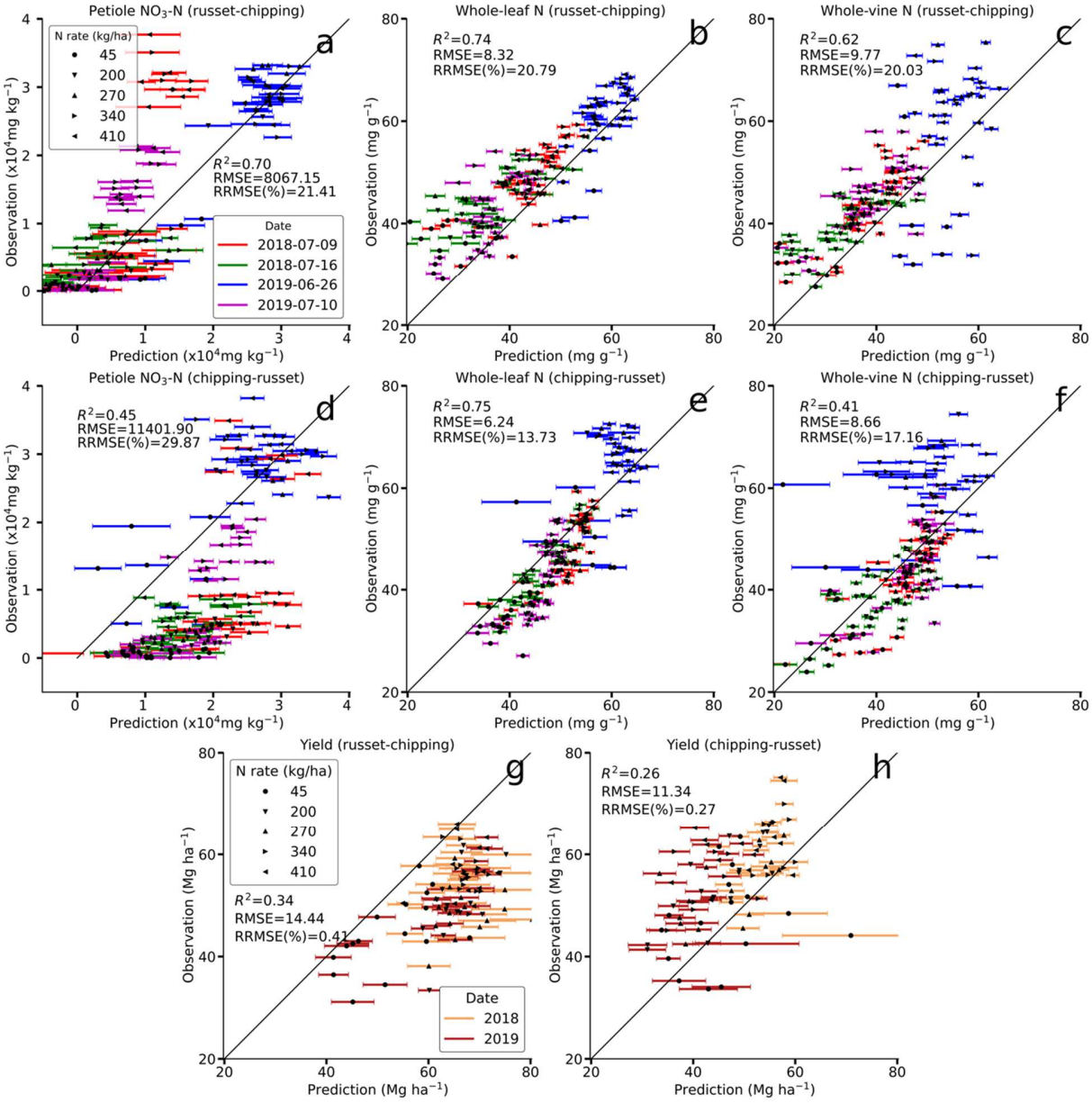
3.5. PLSR modelling validation across potato types

In general, plant tissue N concentration derived from cross-potato type PLSR models showed a moderate-to-strong agreement with field measurements, with a validation R^2 = 0.41-0.74 (Figure 9, a-f). In comparison, cross- potato type models of tuber yield exhibited poor performance (R^2 = 0.26-0.34, Figure 9, g, h).

A systematic bias was observed in model predictions. PLSR models from the russet type tended to overestimate the N concentrations and underestimate the tuber yield for the chipping type while models from the chipping type did the opposite. This bias may be partially explained by the comparison of plant tissue N concentrations and canopy spectra among cultivars in Sections 3.1 and 3.2: N concentrations generally did not show a significant difference among cultivars (Figure 3, b, d, f), while the canopy spectra had notable differences among the two potato types, especially in the NIR region (Figure 4, b, d, f, h).

Although the model coefficients of the two potato types were broadly similar, differences were apparent (Figure 10). For instance, the red and red-edge wavelength region associated with chlorophyll absorption (680-710 nm) was leveraged by the whole vine total N model of the russet type (Figure 10, c). However, it was not indicated as important in the chipping model which more leveraged on the NIR (775-790 nm) region, a region more associated with canopy cover. Similar differences were exhibited in the 750-800, 820-870 and 1550-1600 nm regions for the yield models of two potato types (Figure 10, d).

Interestingly, the whole leaf total N model coefficients were very consistent between two potato types (Figure 10, b). After being rescaled by the magnitude of coefficient values (i.e., $\frac{1}{\alpha_{max}-\alpha_{min}}$, where α_{min} and α_{max} are the minimal and maximal coefficients at all wavelengths), two coefficient curves overlapped with barely any shift. This may explain why the cross-potato type validation of whole leaf total N had the higher R^2 and lower RRMSE than other N concentration variables (Figure 9, b, e).



530

531 Figure 9. Cross-potato type validation results of HySpex PLSR modelling of chipping and russet potatoes.
532 Notes: (1) horizontal bars indicate the uncertainty (±1 standard deviation) in predictions; (2) in each figure
533 title, (A-B) means that type A is used for calibration and type B is used for validation; (3) for N
534 concentrations (petiole NO_3-N, whole leaf total N and whole vine total N), data from all dates are used for
535 modelling; for tuber yield (chipping and russet types), spectral data from 16 July 2018 and 10 July 2019
536 are used for modelling.

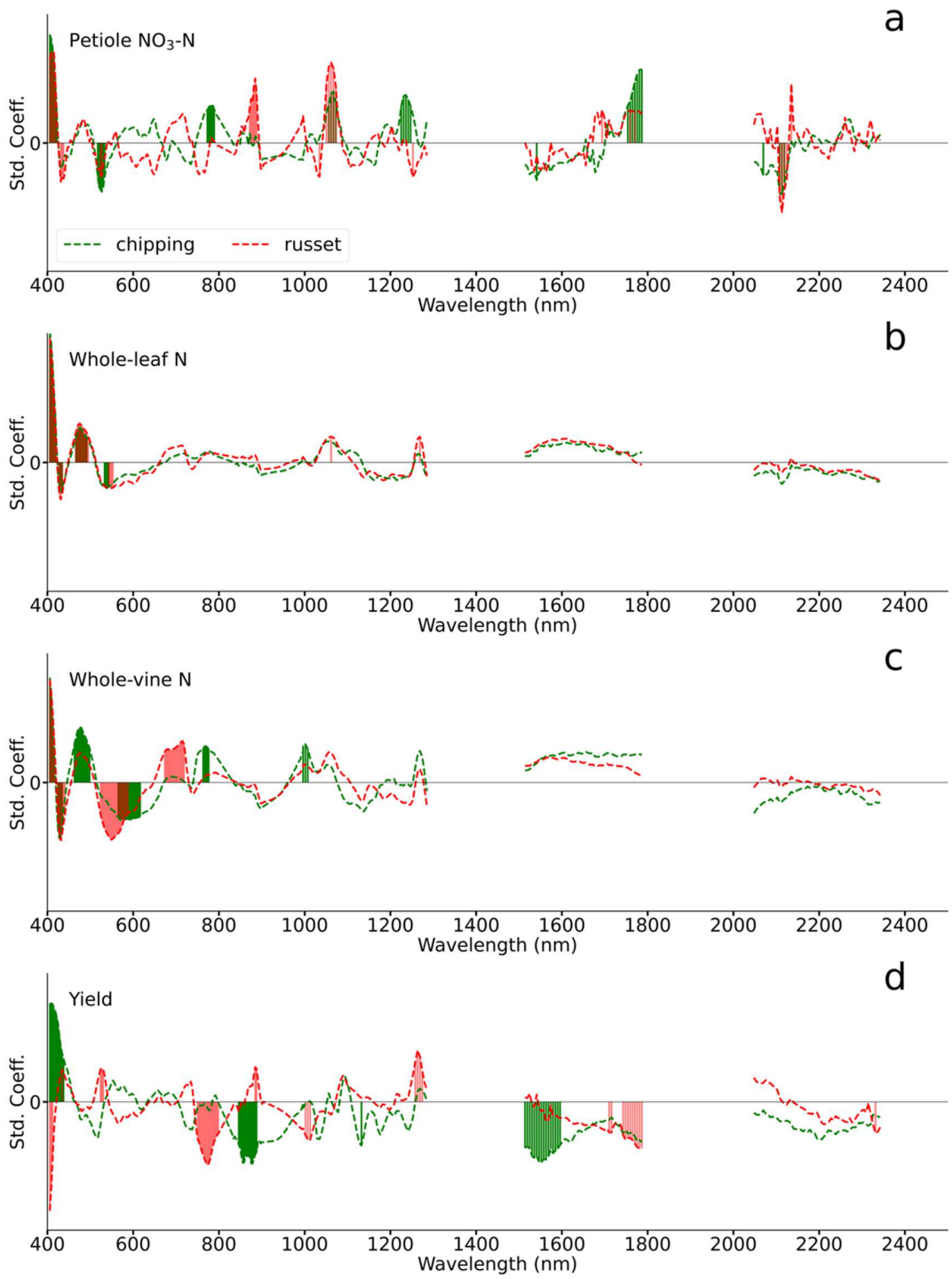


Figure 10. Standardized PLSR model coefficients of two potato types (green: chipping; red: russet). Notes: (1) each coefficient curve is scaled by $\frac{1}{\alpha_{max}-\alpha_{min}}$, where α_{min} and α_{max} are the minimal and maximal coefficients at all wavelengths; (2) vertical bars represent the locations of important wavelengths; (3) full spectrum data were used for modelling.

3.6. PLSR modelling validation across planting seasons

Cross-season predictions of whole leaf and whole vine N had a moderate-to-strong agreement with field measurements, with a validation R^2 = 0.46-0.75 (Figure 11, a-d), with an overprediction bias of 1-2 whole N arising as the major issue for applying 2019 nitrogen models to 2018 imagery. The performance of petiole NO_3-N models was very poor with all R^2s < 0.10 (results not shown in Figure 11). For tuber yield, the validation R^2 of cross-season models varied between 0.07 and 0.34 (Figure 11, e-h).

Systematic biases were observed in model predictions, suggesting that systematic differences may have existed either in the imagery or plant conditions between the two seasons that were not accounted for in a single-season model. This is confirmed in the field data, which showed that the data distributions in the two seasons were distinct (Supplemental Figure S 2). The petiole NO_3-N data of 2018 had a large gap between 10,000 and 25,000 mg kg^{-1}, while the petiole NO_3-N data of 2019 was more continuously distributed (Supplemental Figure S 2, a, b). The whole leaf and whole vine N values of 2018 had a smaller dynamic range (range: 25-62 mg g^{-1}) than 2019 (25-80 mg g^{-1}) (Supplemental Figure S 2, c-f). The applicability of cross-season models is limited when N levels of plants are out the range of the data, or the developmental stages and conditions have not been fully bracketed in the training data.

The limitations imposed by differences in plant metrics across seasons are also illustrated by differences in the PLSR coefficients among models built separately for the two seasons, although less so for whole leaf and whole vine N than tuber yield or petiole NO_3-N (Figure 12). The positions of the most important wavelengths were inconsistent, although in general were in similar positions as other models developed (Figures 8 and 10). For instance, the 1550-1650 nm wavelength region associated with starch and cellulose absorption was identified as important for predicting N concentrations in the PLSR models of 2018, but it was not important in the models of 2019 (Figure 12, a-c). Other wavelengths are consistent, including with the multi-season models (Figure 8 and Figure 10), e.g. in the visible wavelengths and between

1000-1200 nm. This indicates that a range of wavelengths are sensitive to potato nutrient variables, and different wavelengths are emphasized in the spectral models based on conditions specific to the time of measurement.

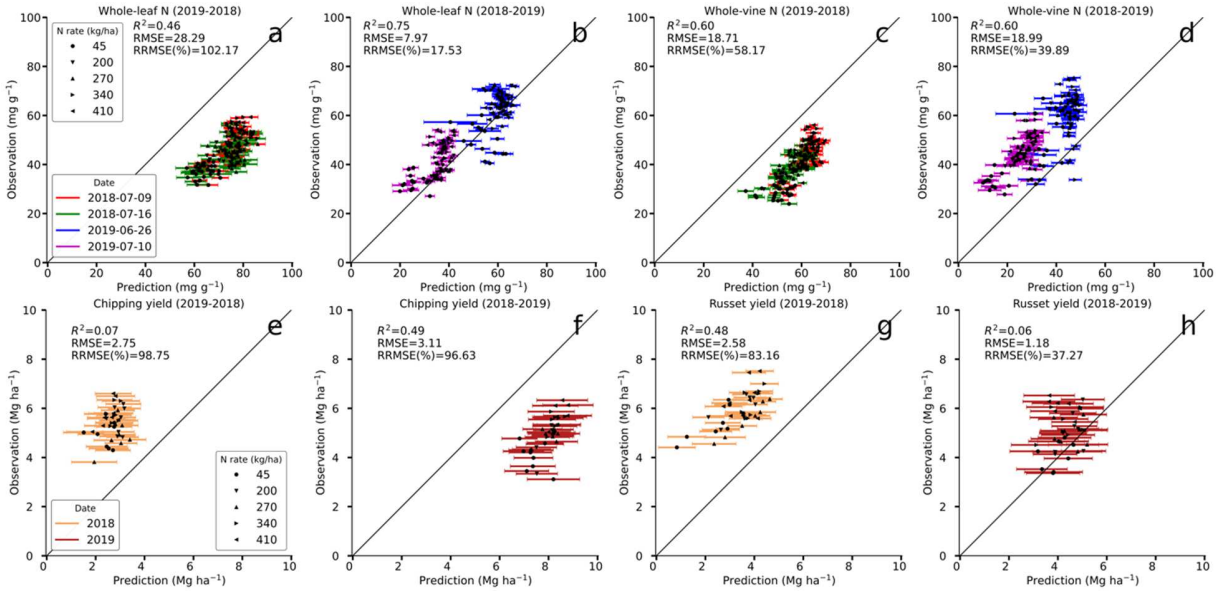
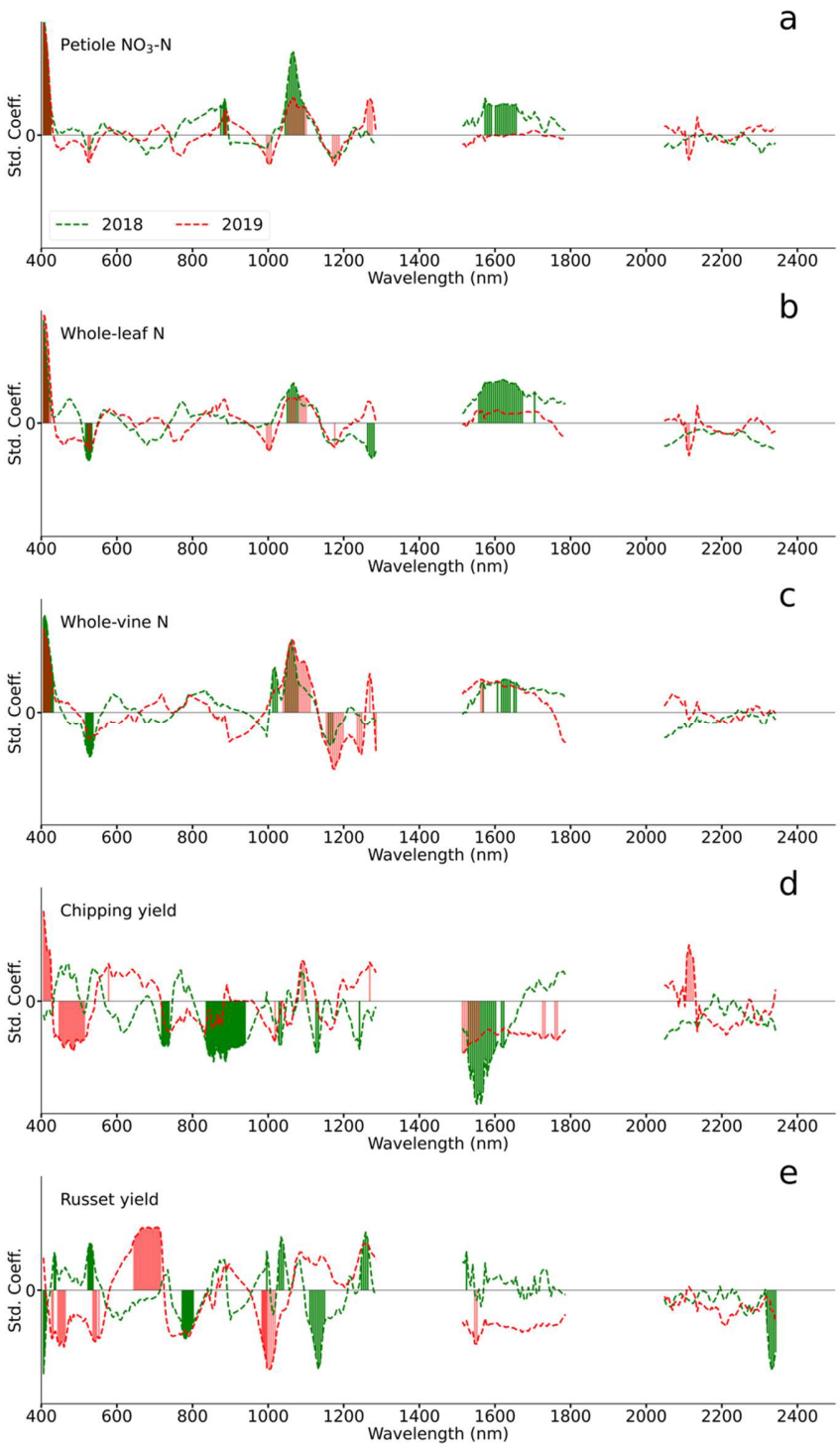


Figure 11. Cross-season validation results of HySpex PLSR modelling (top row: N concentrations; bottom: tuber yield). Notes: (1) horizontal bars indicate the uncertainty (±1 standard deviation) in predictions; (2) in each figure title, (A-B) means season A is used for calibration and season B is used for validation; (3) for N concentration (petiole NO_3-N, whole leaf N and whole vine N), data from all dates are used for modelling; for tuber yield (chipping and russet types), spectral data from 16 July 2018 and 10 July 2019 are used for modelling.

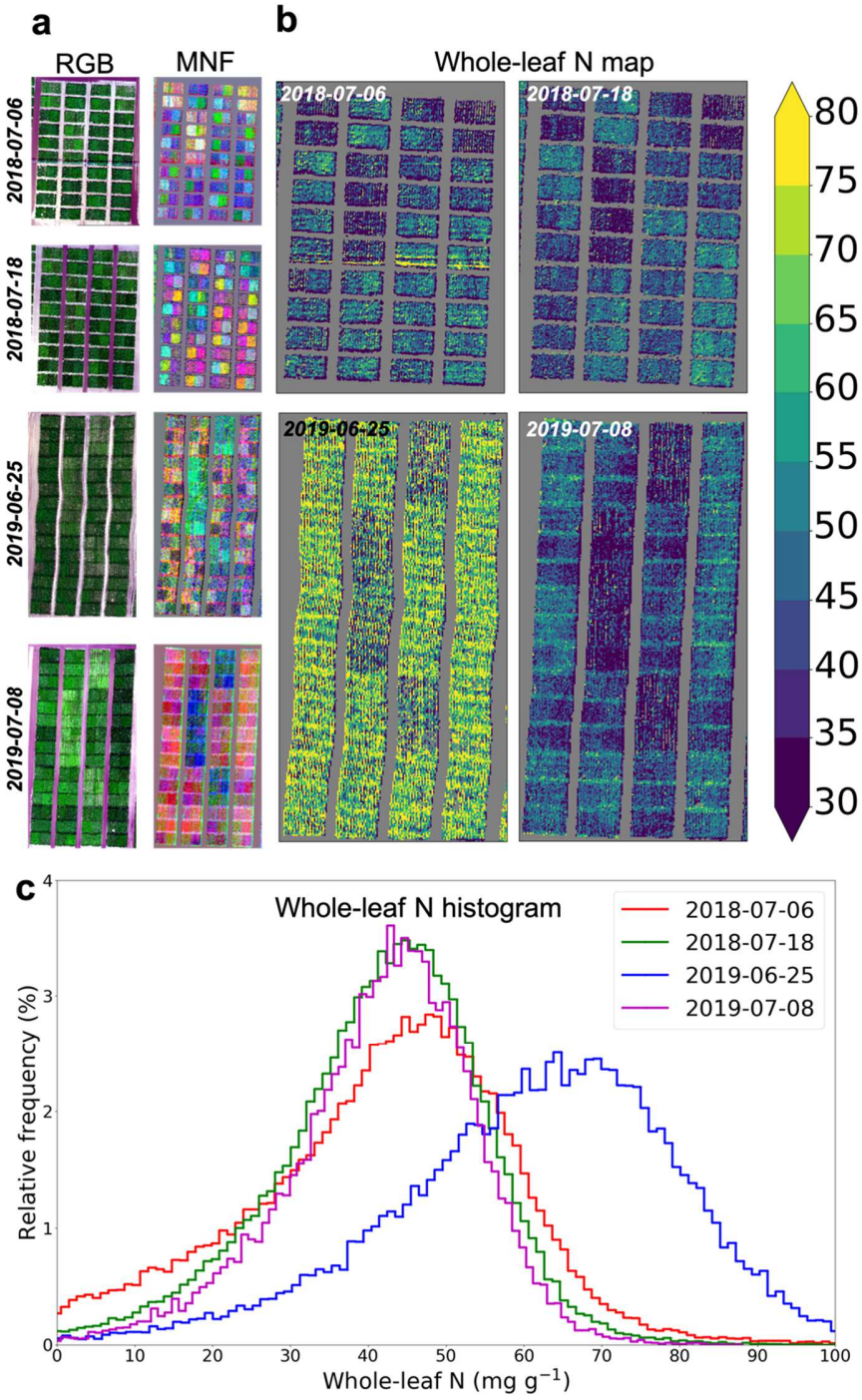


578

579 Figure 12. Standardized PLSR coefficients of each planting season (green: 2018; red: 2019). Notes: (1)
580 each coefficient curve is scaled by $\frac{1}{\alpha_{max}-\alpha_{min}}$, where α_{min} and α_{max} are the minimal and maximal
581 coefficients; (2) vertical bars represent the locations of important wavelengths; (3) full spectrum data are
582 used for modelling.

3.7. N concentration maps

Figure 13 presents the whole leaf N maps created by applying the full-spectrum PLSR model (Section 3.4) to four HySpex images. Apparent differences in whole leaf N maps between dates were observed, with early dates (9 July 2018 and 26 June 2019) having a higher N than later dates (16 July 2018 and 10 July 2019), as is expected over the course of a growing season. On a given date, imaging spectroscopy was able to capture the spectral differences among subplots with different cultivars and N treatments (see the minimum noise fraction (Green et al., 1988) transform images in Figure 13, a). Note that the image of 25 June 2019 retained distortions after geometric corrections due to GPS/IMU issues.



592

593 Figure 13. (a) RGB and MNF (Minimum Noise Fraction transform, used to illustrate hyperspectral data
594 dimensionality) images of four dates; (b) whole leaf N maps of four dates; (c) whole leaf relative frequency
595 histogram. Notes: (1) the first three components are used for displaying the MNF transformed images; (2)
596 the full-spectrum PLSR model in Section 3.4 is used to map whole leaf N; (3) non-vegetation pixels are
597 masked out using NDVI<0.5.

4. Discussion

Our study shows that hyperspectral imaging can be used to map foliar chemistry in potato across a range of varieties, developmental stages, fertilization levels and in different years. This is important because crop management decisions must often be made during the course of a growing season, so that knowledge of ontogenetic variations in plant resource allocation could be used to target fertilization decisions. Our results accurately characterize the seasonal decline in N concentration consistent with a phenological “dilution curve” function that results from the accumulation of biomass as the plant develops and as N is remobilized from the lower canopy to maximize photosynthesis (Lemaire et al., 2008). The utility of hyperspectral remote sensing as a tool for crop and/or nutrient management relies upon accurately depicting processes observed by growers in field settings.

4.1. The implication of SWIR wavelengths for modelling N concentrations

To date, most studies have focused on VNIR (400-1100 nm) wavelengths to assess the N status of potatoes (Botha et al., 2007; Jain et al., 2007; Morier et al., 2015; Ray et al., 2006). These studies simply relied on the chlorophyll-nitrogen relationship (Gianquinto et al., 2004; Olivier et al., 2006; Wu et al., 2007). That is, high N rates tend to promote higher LAI and thus stronger chlorophyll absorption and consequently lower VIS but higher NIR reflectance (Figure 4, a, c, e, f). However, chlorophyll should not be considered as a reliable indicator of leaf N concentration, as much of the N in foliage is in the enzyme rubisco, which is responsible for the uptake of CO_2 and varies largely independently of leaf chlorophyll content.

The VNIR-only is attractive when a more expensive SWIR instrument is not available (Bohman et al., 2019; Franceschini et al., 2017). Yet, an extensive literature has emerged in recent years demonstrating the power of full spectrum VSWIR (400-2500 nm) imaging for crop assessments in general, and potatoes in particular, as the SWIR wavelengths have important features related to nutrients and overall plant physiology (Gold et al., 2020, 2019).

623 Our OLSR (Figure 5, dry matter VIs) and PLSR (Figure 6, g-i and Figure 7, e, f) results further
624 confirmed the importance of SWIR wavelengths. Even though the variation of reflectance was
625 small among N rates (Figure 4, a, c, e, f), SWIR wavelengths related to dry matter content and
626 known nitrogen/protein absorption features (Table 2, dry matter VIs; Figure 8, red bars) were
627 still found to be diagnostic of differences in plant N status. Moreover, SWIR-only may be the
628 best option for situations where haze or other atmospheric effects reduce the utility of VNIR data,
629 as is common in the summer. A SWIR-only model may be expected to be relatively robust
630 across a range of crops and use cases for nitrogen variables, as the SWIR wavelengths are
631 those that are most strongly related to nitrogen absorption features associated with proteins
632 (Berger et al., 2020b; Curran, 1989), independent of potential confounding variables such as
633 chlorophyll and canopy structure. These findings provide strong implications for using data from
634 current and forthcoming spaceborne VSWIR imaging spectrometers (e.g., EnMap, PRISMA,
635 CHIME, SBG).

636 **4.2. The influence of sampling dates on modelling N concentrations**

637 Differences in field data distribution among growth stages (e.g., Figure 3) and planting seasons
638 (e.g., Supplementary Figure S 2) are major considerations for using imaging spectroscopy as a
639 management tool, and broader application will require sufficient representation of relevant
640 conditions, cultivars and growth stages. In our study, the data used for modelling N
641 concentrations were collected only in the early- and mid-tuber bulking stages of two growing
642 seasons (Sections 2.1 and 2.2). The inter-annual analysis in Section 3.6 points to the
643 importance of comprehensively characterizing potato conditions across a range of
644 developmental stages (e.g., at emergence, hilling and tuber maturity) and environmental
645 conditions in multiple seasons to build robust and generalizable models. However, some caution
646 is called for when doing so. First, the effect of mixed pixels on spectral reflectance needs to be
647 considered. In this study, HySpex images were taken at early- and mid-tube bulking stages
648 when plant canopies were close to 100% closed, and at a fine spatial resolution (0.15-0.25 m)

649 with which bare soil or shadow pixels could be easily removed (Section 2.2). Therefore, all
650 pixels used for averaging the spectra within each plot could be viewed as 'pure' vegetation
651 pixels (Section 2.2). However, the confounding effects of bare soil and senesced vegetation
652 would be issues for spectra collected at early or late growth stages, or at coarse spatial
653 resolutions (Asner, 1998; Morier et al., 2015). For instance, in a recent study by Wang et al.
654 (2019), the estimation of foliar biochemistries for plots with low vegetation cover was found to
655 have high uncertainties.
656 Second, it has been suggested that field data obtained at early growth stages with small
657 dynamic ranges may not be suitable for calibrating models (Botha et al., 2007; Franceschini et
658 al., 2017; Nigon, 2012). In this study, the N concentration for 26 June 2019 produced poor
659 model predictions (Figure 5; Figure 6, blue bars; Figure 9, blue bars). When assessed on its
660 own, this time point had insufficient dynamic range, as field plots had small variations in N
661 concentration (Figure 3, 2019-06-26) as well as canopy spectral reflectance (Figure 4, 2019-06-
662 26).
663 Finally, limitations imposed by plant characteristics can be exacerbated by inconsistencies in
664 measured canopy spectra acquired on different dates, irrespective of plant status. In this study,
665 we implemented consistent atmospheric corrections, BRDF corrections and spectral
666 normalizations to reduce the spectral discrepancy across dates (Section 2.2). However, it is also
667 likely that effects of imperfect image corrections propagate to the subsequent analyses (here,
668 various types of PLSR modelling). Atmospheric correction models can never perfectly
669 parameterize atmospheric conditions at the time of overflights, which can result in systematic
670 differences in spectral data across dates. This is the primary reason why we vector-normalized
671 the data prior to analysis, but if discrepancies are unequal among spectral regions (e.g., haze
672 affecting the visible but not SWIR), in which case vector normalization may have no effect or
673 even reduce comparability among scenes from different dates. To address this, ground pseudo-
674 invariant features (e.g., calibration panels, bare ground or tarps) with consistent reflectance can

675 be used to normalize/calibrate images (Shrestha et al., 2019; Yang et al., 2017), although such
676 an approach is not practical in broad application and across multiple sites and dates.
677 New processing approaches are now emerging for normalizing spectra. For example, Tan et al.
678 (2020) proposed a radiance correction method (as opposed to reflectance) to remove the bi-
679 directional reflectance and atmospheric attenuation effects caused by sun-ground-sensor
680 geometry angles and flight altitudes. Their results showed that this method could effectively
681 remove the radiance mismatch between image dates and among flight lines on the same date.
682 As well, Thompson et al. (2018) introduced a new atmospheric correction approach based on
683 optimal estimation that works well in variable landscapes over complex cover types. While these
684 approaches are not yet available for broad implementation, in the near future they should offer
685 opportunities to develop more consistent reflectance products that will subsequently enable
686 consistent spectral retrievals through time.

687 **4.3. The challenges in modelling petiole NO_3-N**

688 Among the three measures of plant N concentration, whole leaf N was best predicted (Figure 5
689 and Figure 6, map shown in Figure 13). This has been also found in previous studies (Cohen et
690 al., 2010; Nigon, 2012; Nigon et al., 2014). The result is not surprising because plant leaves are
691 the dominant contributor to canopy reflectance.
692 Detection of properties that are not directly sensed, such as petiole NO_3-N, likely leverages
693 correlations with whole leaf N. Petiole NO_3-N would have been expected to perform most poorly;
694 although it is correlated with leaf N (here, r = 0.76-0.88), petioles make a negligent contribution
695 to the sensed spectra. Also, as noted in Section 3.1, the petiole data for some dates (e.g., 16
696 July 2018, Figure 3) were highly variable. Although whole vine N is correlated with whole leaf N
697 (here, r = 0.70-0.80), it is also affected by other plant organs such as stems and reproductive
698 structures.
699 Petiole NO_3-N is of special interest to potato growers because of its use as a diagnostic
700 measure of nutrient availability. At best, the imagery reported here could detect high and low

701 petiole NO_3-N levels but not finer gradations. Aside from petioles being a small percentage of
702 the observed canopy, another issue for detecting petiole NO_3-N lies in the timing of fertilizer
703 application, image acquisition and tissue sampling. For instance, plants were fertilized on 2 July,
704 imaged on 6 July and tissue samples were collected on 9 July (Figure 1, c). Tissue analyses
705 generally should occur four days or longer following fertilization to allow for uptake and
706 assimilation of nutrients and the stabilization of labile pools of nitrate in petioles. In contrast, leaf
707 and plant N concentrations are integrated metrics of seasonal uptake and use. Across all
708 samples, the petiole NO_3-N data exhibited a large gap between 10000 and 25000 mg kg^{-1}
709 (Figure 3, a), with all of the high points being from rates 340 and 410 kg ha^{-1} imaged on 06 July
710 and sampled on 09 July, after receiving a N application of 70 kg ha^{-1} on 02 July (Figure 1, c).
711 Although our image was acquired four days (06 July) after the N application, it is possible that
712 NO_3-N on 06 July in some plants may not have equilibrated to the level observed on 09 July.
713 We also noticed that the PLSR model for petiole NO_3-N produced some negative predictions at
714 low concentrations (Figure 6, a, d, g), which was also observed by Cohen et al. (2010).
715 Negative predictions are not uncommon with data sets whose lower limits are very small
716 positive numbers.

717 **4.4. The potential of other statistical or physically-based models**

718 In this study, we evaluated the performance of PLSR to predict N concentration from imaging
719 spectra. Other data-driven techniques such as Gaussian process (GP), random forest (RF) and
720 artificial neural network (ANN) models may be worthy of future investigations (Berger et al.,
721 2020a; Verrelst et al., 2016a). Compared to PLSR, these methods may capture the nonlinear
722 relationships between spectral signatures and nitrogen. While some comparative studies
723 demonstrate higher prediction accuracy than PLSR (Yao et al., 2015; Zhou et al., 2018), others
724 show that they are comparable (Wang et al., 2019) and the resulting models from machine
725 learning approaches are less interpretable than PLSR, which provides a signed coefficient that

726 can be interpreted directly with respect to known absorption features associated with plant
727 physiology or biochemistry.
728 Recent development in radiative transfer models (RTMs) provides new insights to estimating
729 foliar N via physically-based models (Berger et al., 2020b; Féret et al., 2020; Wang et al., 2015).
730 For example, Féret et al. (2020) recalibrated the PROSPECT leaf radiative transfer model. In
731 the new PROSPECT version, named PROSPECT-PRO, leaf mass per area (LMA) was
732 replaced by nitrogen-containing proteins and carbon-based constituents (CBC, e.g., lignin,
733 cellulose, hemicellulose, sugar and starch) as new leaf input constituents. The model inversion
734 of PROSPECT-PRO showed that leaf nitrogen can be quantified by estimating leaf protein
735 content with a reasonable accuracy (validation R^2=0.45-0.79). Based on this work at the leaf
736 level, Berger et al. (2020a) coupled PROSPECT-PRO with the canopy radiative transfer model,
737 4SAIL (Verhoef et al., 2007), to estimate foliar N from canopy reflectance. Compared to
738 statistically-based models, physically-based models are more robust and transferable (Berger et
739 al., 2020b) since they can simulate the canopy reflectance under different planting conditions.
740 The disadvantage of physically-based models is the problem of ill-posedness: the number of
741 unknown biophysical variables is typically much larger than the number of reflectance
742 observations (Verrelst et al., 2015). To estimate foliar N, crop crown characteristics related to
743 canopy architecture and physiognomy, such as leaf area index, need to be provided as a prior
744 information for a given flight acquisition.

745 **4.5. The efficiency of real time foliar nitrogen mapping**

746 Real-time nutrient mapping is key for agriculture applications. In our study, the pre-processing of
747 HySpex images took most of the time of nutrient mapping. In general, a HySpex VNIR (~4000
748 rows by 1800 columns) and SWIR (~2000 by 384) image pair needed about 3-4 hours for data
749 preprocessing on a desktop workstation using our HyPro workflow, although commercial
750 packages with similar functionality may perform faster due to more efficient interpolation of
751 atmospheric look-up-tables. Implementation of PLS algorithms to make maps on pre-processed

752 images is fast (< 5 minutes per flightline), but parallel processing is likely to be necessary to
753 ensure real-time nutrient mapping for large numbers of flightlines that require pre-processing
754 using this approach. In the future, an emulator based on machine learning algorithms may
755 effectively substitute atmospheric RTMs to nearly instantly simulate radiative transfer for
756 atmospheric correction (Shi et al., 2019; Verrelst et al., 2016b).

757 **5. Conclusion**

758 We investigated the potential of OLSR and PLSR to predict four potato traits from aerial
759 hyperspectral remote sensing: petiole NO_3-N, leaf total N, canopy total N and tuber yield. Our
760 study comprehensively addressed the capacity of chemometric models to estimate important
761 variables for nutrient management in potatoes, addressing a range of considerations that will be
762 important to growers including seasonal growing conditions, growth stages and cultivars.
763 Our results provide evidence that generalizable models are possible, as long as they have been
764 built using calibration and validation ground measurements that capture the dynamic range of
765 plant growth conditions. Robust, multi-season models necessitate training measurements from
766 all developmental stages of interest and for multiple growing seasons with variable weather
767 conditions. Such models may prove useful for real-time monitoring of plant N status and nutrient
768 management decision making.

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List of Figure Captions

Figure 1. A) field experimental design (2018: 80 plots = 5 N rates × 4 cultivars × 4 replicates; 2019: 120 = 5 × 6 × 4; black dashed-line boxes show a whole plot containing four or six cultivar subplots). B) the Hancock agricultural research station (HARS) in Wisconsin, USA. C) N application and timing for each of five N rates. D) list of potato cultivars grouped into two potato types (or market segments).

Figure 2. Repeated k-fold cross-validation procedure.

Figure 3. Comparison of potato N concentration (including petiole NO3-N (a, b), whole leaf total N (c, d), whole vine total N (e, f) and tuber yield (g, h) among five N rates (left column) and four potato cultivars (right column).

Figure 4. Comparison of mean reflectance (0-100%) among five N rates (left column) and four potato cultivars (right column). Mean reflectance was calculated by averaging the spectra of plots with the same N rate or potato cultivar.

Figure 5. OLSR modelling validation results (left axis, red and purple symbols: R2; right axis, blue and gold symbols: relative RMSE) of N concentrations including petiole NO3-N, whole leaf N and whole vine N, and tuber yield for chipping (solid lines) and russet (dash lines) types using different VIs in Table 2. Background colors represent different groups of VIs. Green: chlorophyll-sensitive; Cyan: water; Orange: dry matter.

Figure 6. HySpex PLSR modelling validation results of petiole NO3-N (left column), whole leaf N (middle column), and whole vine N (right column). Notes: (1) horizontal bars indicate the uncertainty (±1 standard deviation) in predictions; (2) full-spectrum: 400-2500 nm (top row); VNIR-spectrum: 400-1300 nm (middle row); SWIR-spectrum: 1400-2500 nm (bottom row).

Figure 7. HySpex PLSR modelling validation results of tuber yield over two years (left column: chipping type; right: russet type). Notes: (1) horizontal bars indicate the uncertainty (±1 standard deviation) in predictions; (2) full-spectrum: 400-2500 nm; VNIR-spectrum: 400-1300 nm; SWIR-spectrum: 1400-2500 nm; (3) spectra from 16 July 2018 and 10 July 2019 are used for yield modelling.

Figure 8. Standardized PLSR coefficients using different spectral wavelength regions (full-spectrum in blue: 400-2500nm; VNIR-spectrum in green: 400-1300nm; SWIR-spectrum in red: 1400-2500nm). Notes: (1) each coefficient curve is scaled by 1/(α_"max" -α_"min"), where α_"min" and α_"max" are the minimal and maximal coefficients; (2) vertical bars represent the locations of wavelengths important for predicting potato variables.

Figure 9. Cross-potato type validation results of HySpex PLSR modelling of chipping and russet potatoes. Notes: (1) horizontal bars indicate the uncertainty (±1 standard deviation) in predictions; (2) in each figure title, (A-B) means that type A is used for calibration and type B is used for validation; (3) for N concentrations (petiole NO3-N, whole leaf total N and whole vine total N), data from all dates are used for modelling; for tuber yield (chipping and russet types), spectral data from 16 July 2018 and 10 July 2019 are used for modelling.

Figure 10. Standardized PLSR model coefficients of two potato types (green: chipping; red: russet). Notes: (1) each coefficient curve is scaled by 1/(α_"max" -α_"min"), where α_"min" and α_"max" are the minimal and maximal coefficients at all wavelengths; (2) vertical bars represent the locations of important wavelengths; (3) full spectrum data were used for modelling.

Figure 11. Cross-season validation results of HySpex PLSR modelling (top row: N concentrations; bottom: tuber yield). Notes: (1) horizontal bars indicate the uncertainty (±1 standard deviation) in predictions; (2) in each figure title, (A-B) means season A is used for calibration and season B is used for validation; (3) for N concentration (petiole NO3-N, whole leaf N and whole vine N), data from all dates are used for modelling; for tuber yield (chipping and russet types), spectral data from 16 July 2018 and 10 July 2019 are used for modelling.

Figure 12. Standardized PLSR coefficients of each planting season (green: 2018; red: 2019). Notes: (1) each coefficient curve is scaled by 1/(α_"max" -α_"min"), where α_"min" and α_"max" are the minimal and maximal coefficients; (2) vertical bars represent the locations of important wavelengths; (3) full spectrum data are used for modelling.

Figure 13. (a) RGB and MNF (Minimum Noise Fraction transform, used to illustrate hyperspectral data dimensionality) images of four dates; (b) whole leaf N maps of four dates; (c) whole leaf relative frequency histogram. Notes: (1) the first three components are used for displaying the MNF transformed images; (2) the full-spectrum PLSR model in Section 3.4 is used to map whole leaf N; (3) non-vegetation pixels are masked out using NDVI<0.5.